

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: DskEth-mwRev-Cf.sil etc.

Creation date: 10-Feb-82 9:08:47

For: Spruce

50 total sheets = 49 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Character code 57b not found in font Gates32.

Character w/ code 105b, font Helvetica7I would lie off the page.

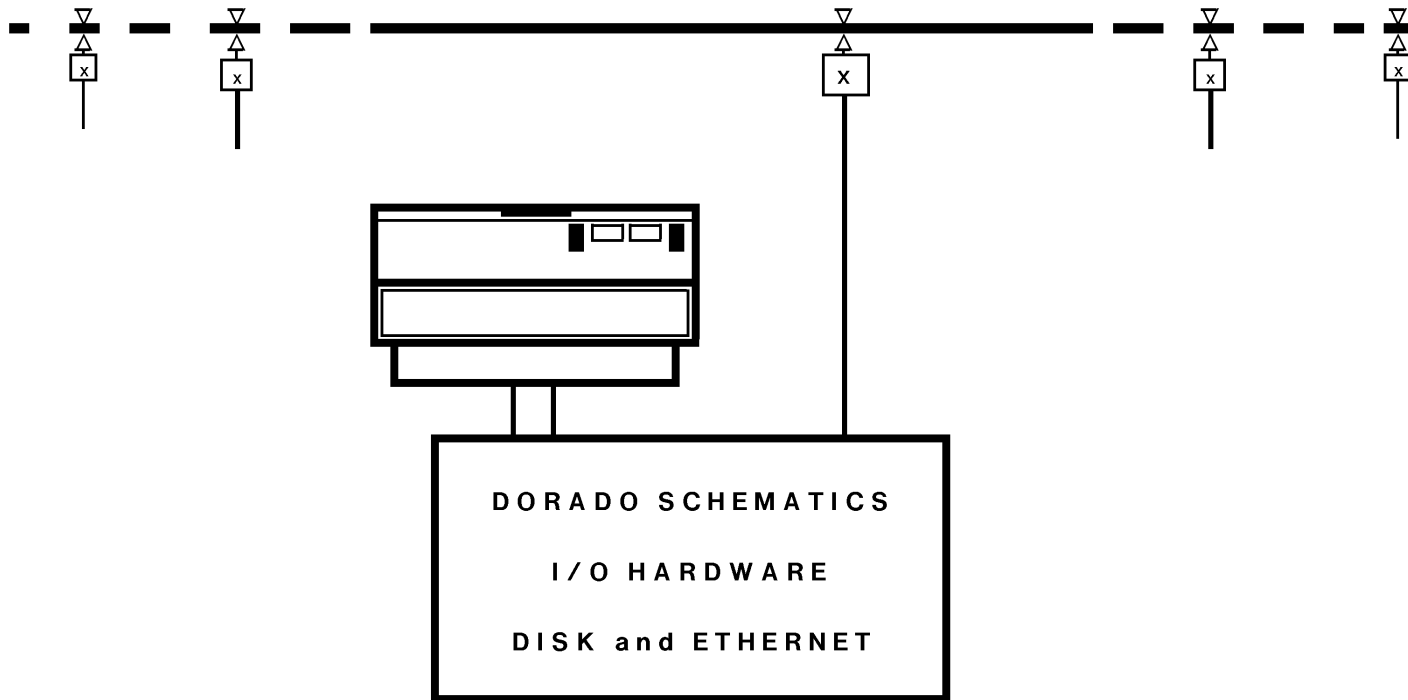
Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

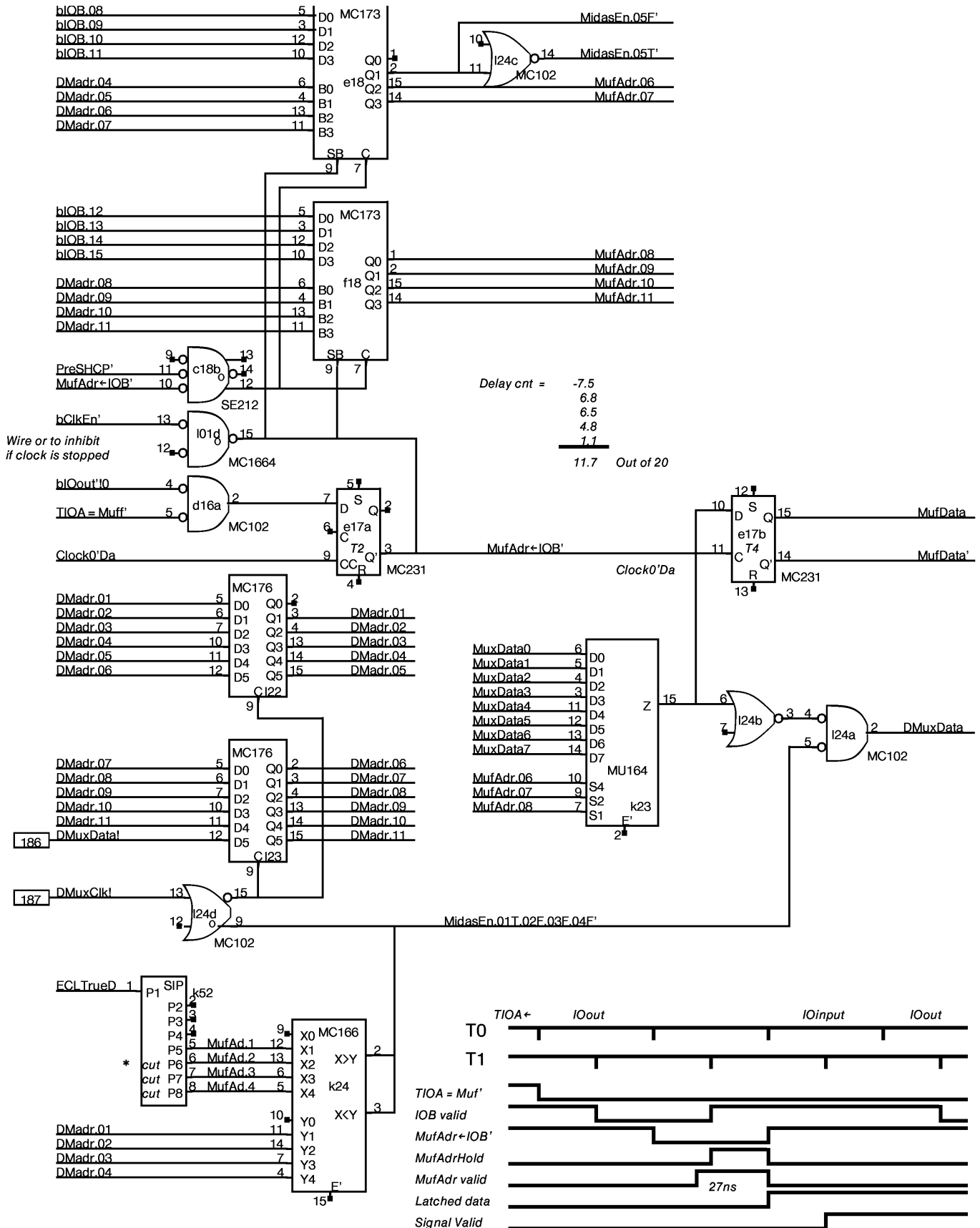
Character code 117b not found in font Gates32.

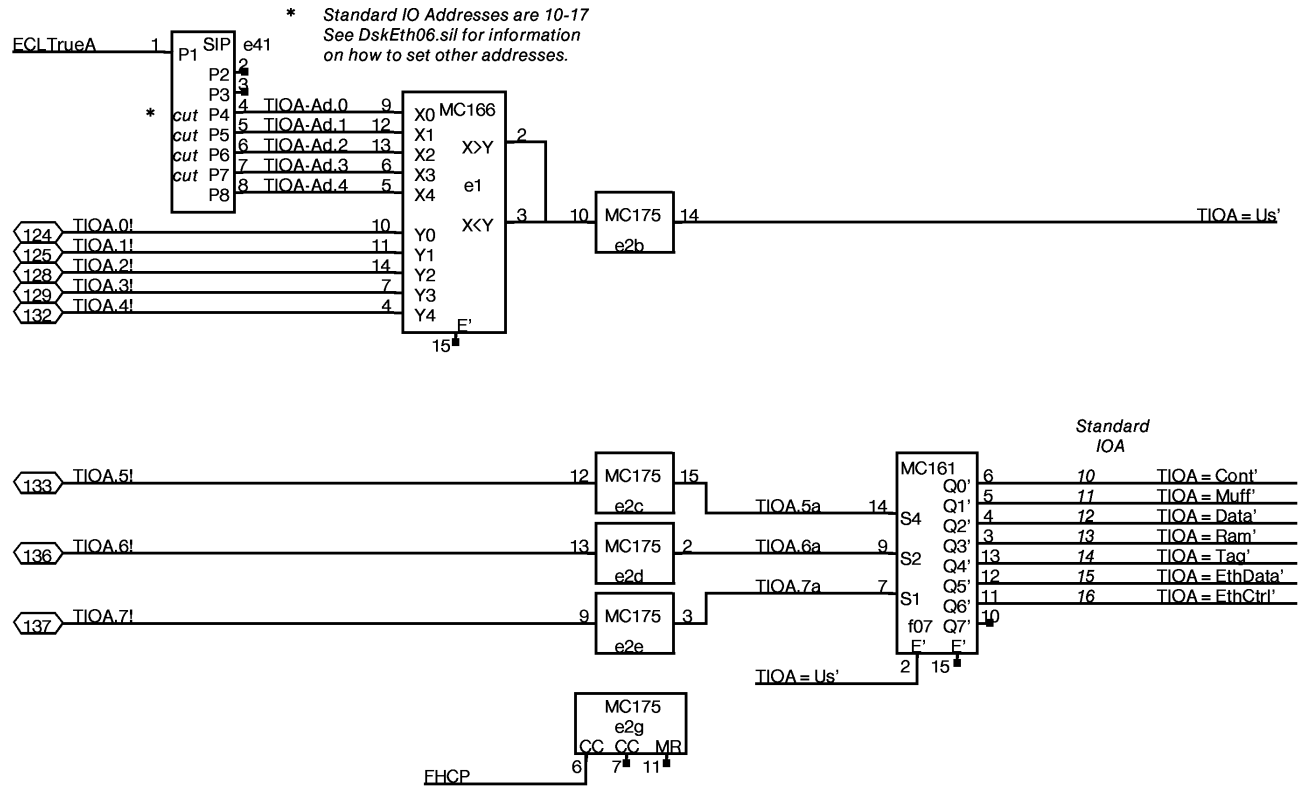
Character code 157b not found in font Gates32.

Character code 117b not found in font Gates32.



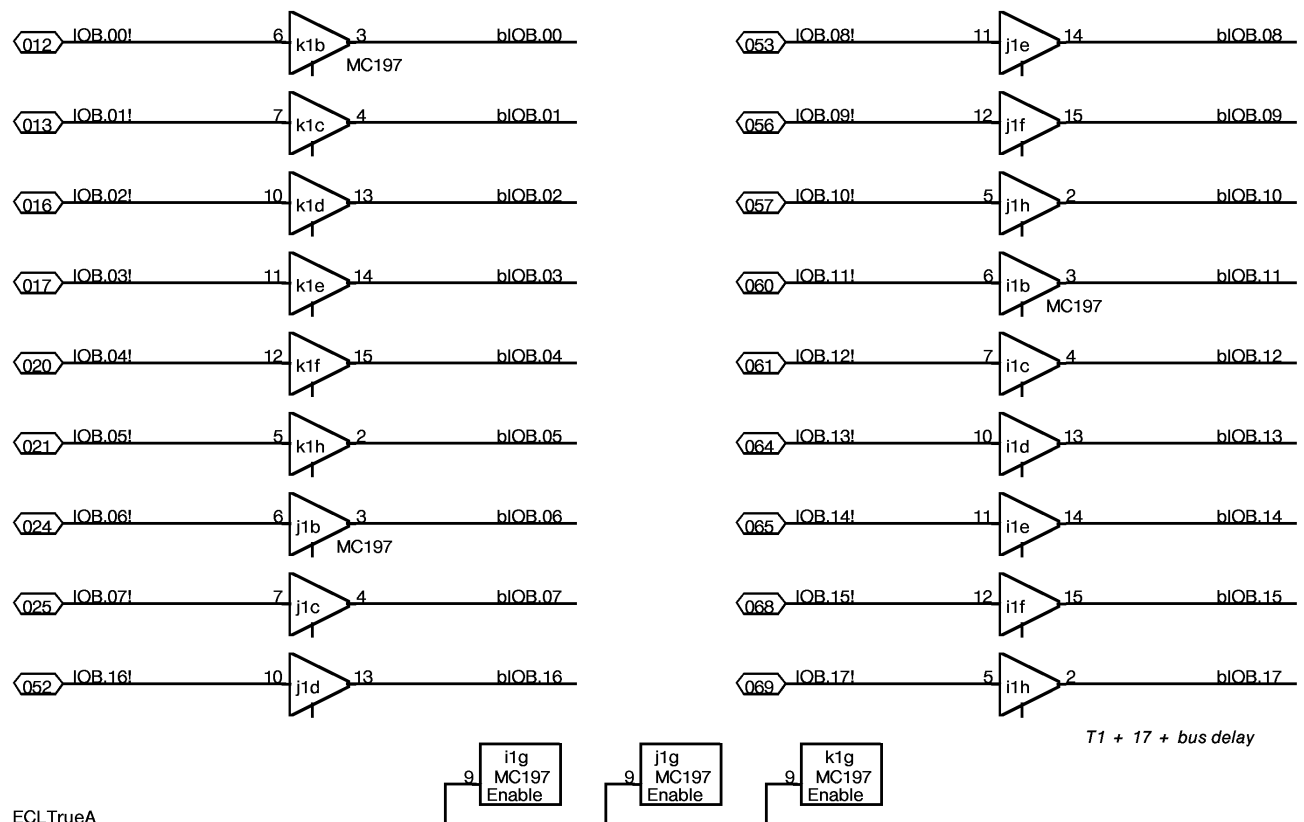
1) Drawings of common logic	01
Midas Muffler Control	01
IOA and IOB	02
Clocks and Temp sense	04
Layout	05
Configuration	06
2) Drawings for TriconD disk Controller	07
State Control Register	08
Format Ram, Counter and Proms	09
Tag Register	10
Disk Drive Control	11
FIFO	13
Error Correction Shift Register	15
Task Wake-Up and IOB parity check	16
Mufflers	18
Clocks	19
I/O pins and Termination	20
Timing Diagram	21
Cable Assembly Drawings	22
3) Drawings for Ethernet Controller	23
Receiver	24
Transmitter	29
Test Logic	34
Clocks	35
Next Bus and IOattention	36
Mufflers	37
Cable I/O and termination	38
Timing Diagrams	40



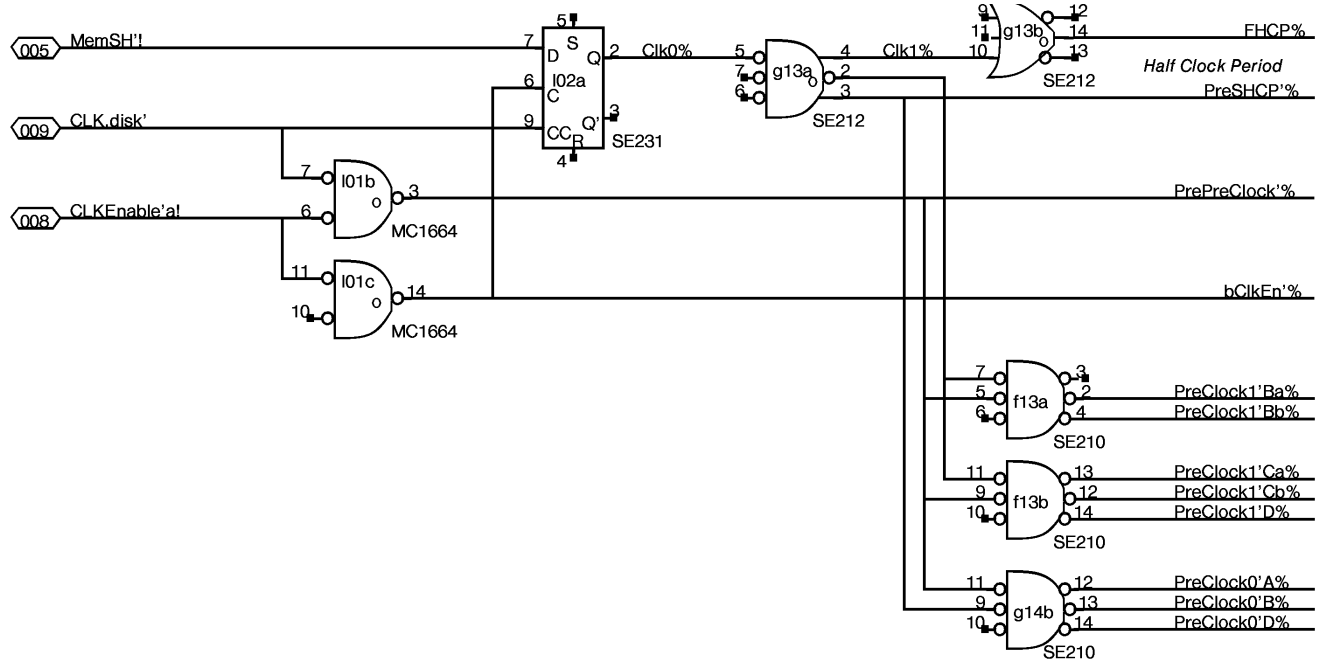


IO Address decoding

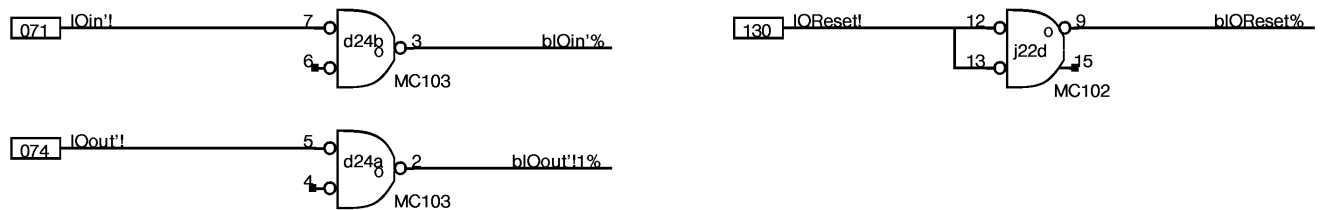
IOB receivers



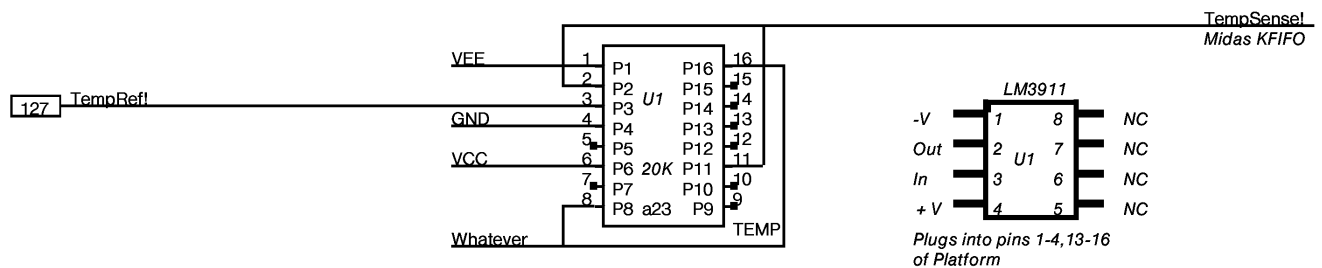
NOTE: IOB data received by this board is tested as part of the disk controller (Pg 18)
Both the disk and the Ethernet check the data parity after it has gone
through the respective FIFO's.



Common Clock circuitry



Common IO circuitry

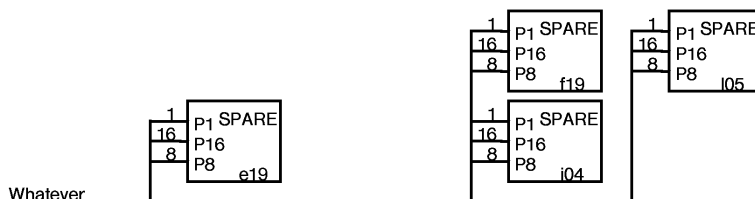


Temperature Sensor

Power connections are as follows:

Stitch-Weld: 1 & 16 are GND
8 is -5

Multiwire: 16 is GND
8 is -5
except in locations
a18, g03, and g16
which are uncommitted



Spare Socket locations for Multiwire

XEROX	Project	Drawing	File	Designer	Rev	Date	Page
PARC	Dorado	Clocks & IO Signals	DskEth04.sil	Bates/Boggs	Cf	2/09/82	04



31 chips common to Disk & Ether
 137 chips specific to Disk
 111 chips specific to Ether
 9 spare chip positions
 288 total chip positions

* Standard addresses are 2000-2177.

	Muff Addr	P5	P6	P7	P8
	0000-0177	cut	cut	cut	cut
	0200-0377	cut	cut	cut	
	0400-0577	cut	cut		cut
	0600-0777	cut	cut		
	1000-1177	cut		cut	cut
	1200-1377	cut		cut	
	1400-1577	cut			cut
	1600-1777	cut			
*	2000-2177		cut	cut	cut
	2200-2377		cut	cut	
	2400-2577		cut		cut
	2600-2777		cut		
	3000-3177			cut	cut
	3200-3377			cut	
	3400-3577				cut
	3600-3777				

* Standard tasks are 6 & 7.

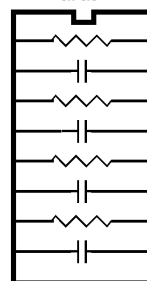
	Tasks	P6	P7	P8
	2 & 3			cut
	4 & 5		cut	
*	6 & 7		cut	cut
	8 & 9	cut		
	10 & 11	cut		cut
	12 & 13	cut	cut	
	14 & 15	cut	cut	cut

* Standard addresses are 10-17.

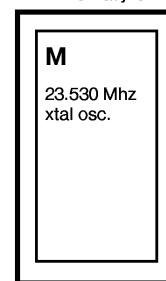
	IOA	P4	P5	P6	P7	P8
	000-007	cut	cut	cut	cut	cut
*	010-017	cut	cut	cut	cut	
	020-027	cut	cut	cut		cut
	030-037	cut	cut	cut		
	040-047	cut	cut		cut	cut
	050-057	cut	cut		cut	
	060-067	cut	cut			cut
	070-077	cut	cut			
	100-107	cut		cut	cut	cut
	110-117	cut		cut	cut	
	120-127	cut		cut		cut
	130-137	cut		cut		
	140-147	cut			cut	cut
	150-157	cut			cut	
	160-167	cut				cut
	170-177	cut				
	200-207		cut	cut	cut	cut
	210-217		cut	cut	cut	
	220-227		cut	cut		cut
	230-237		cut	cut	cut	
	240-247		cut		cut	cut
	250-257		cut			
	260-267		cut			cut
	270-277		cut			
	300-307			cut	cut	cut
	310-317			cut	cut	
	320-327			cut		cut
	330-337			cut		
	340-347				cut	cut
	350-357				cut	
	360-367					cut
	370-377					

Name	Type	#	Location		
DskEth					
Disk					
DiskRead	SG139	2	a20	a21	
DiskTag	MC149	1	d21		
DiskUnits	S288	1	d05		
DiskFifo	MC149	1	b14		
Ether					
EtherPD	MC149	1	h22		
EtherRcvr	MC149	3	h09	h10	h11
EtherFifo	MC149	2	l10	l15	
EtherXmtr	MC149	3	h14	h15	h16

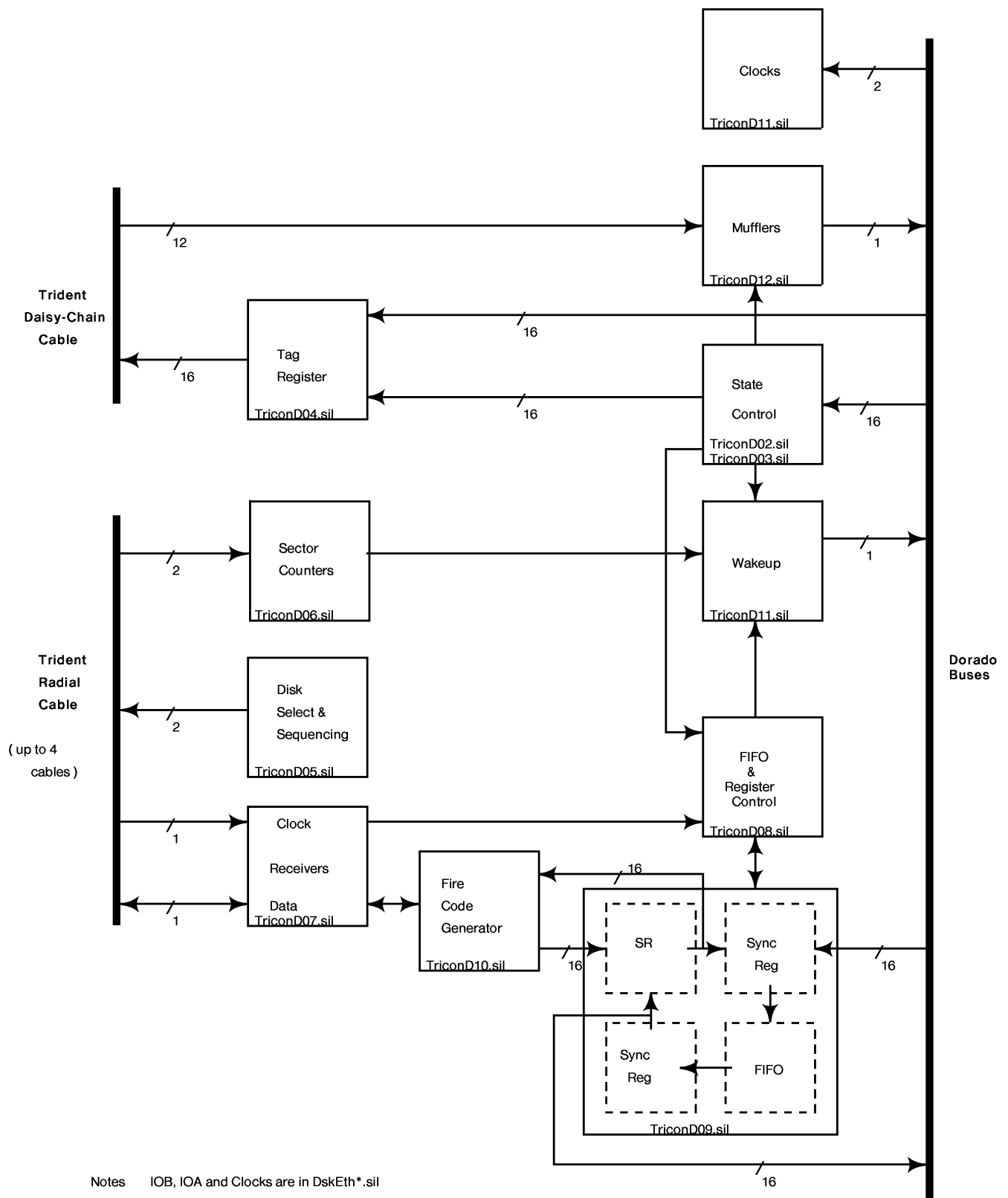
PLAT at a04

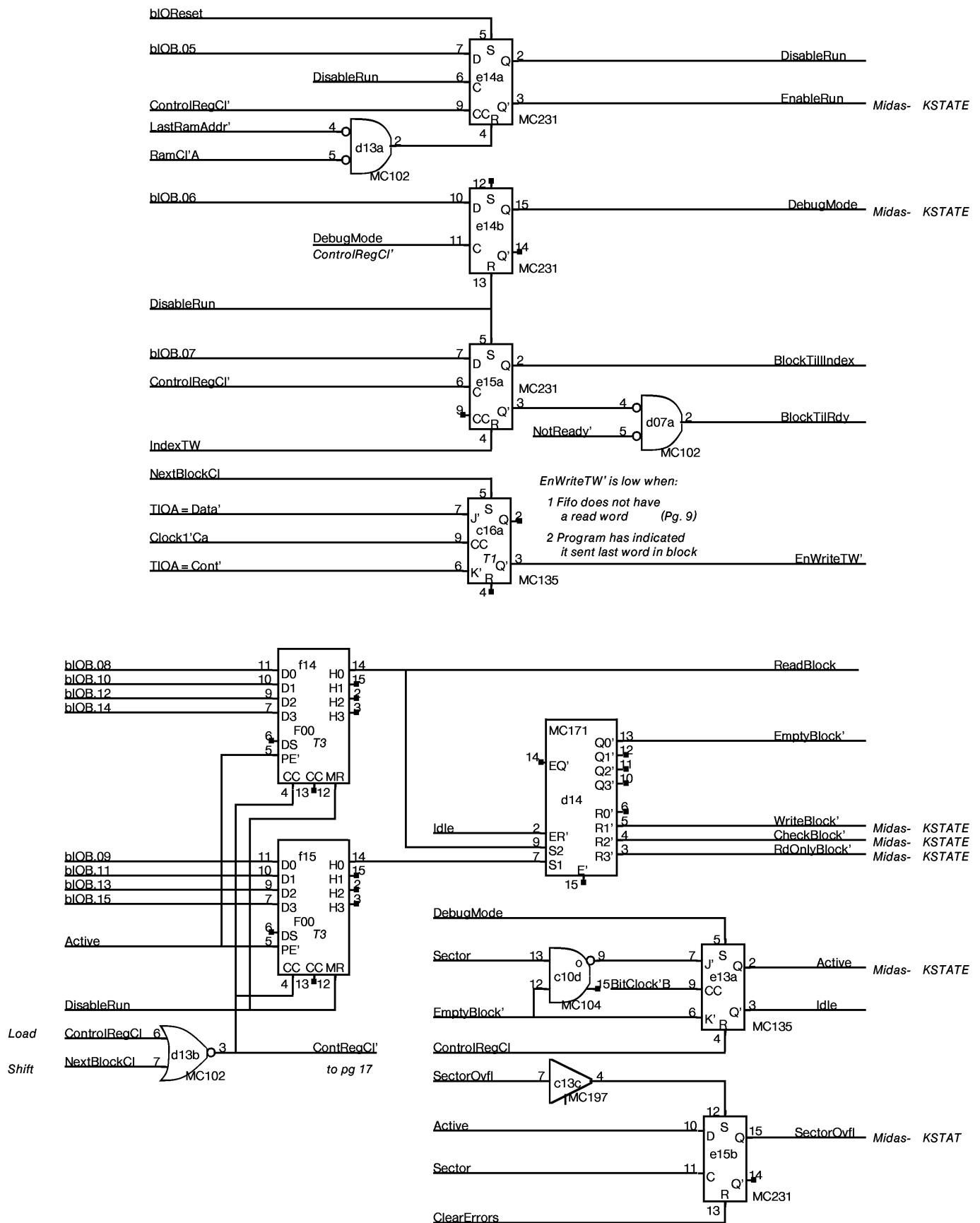


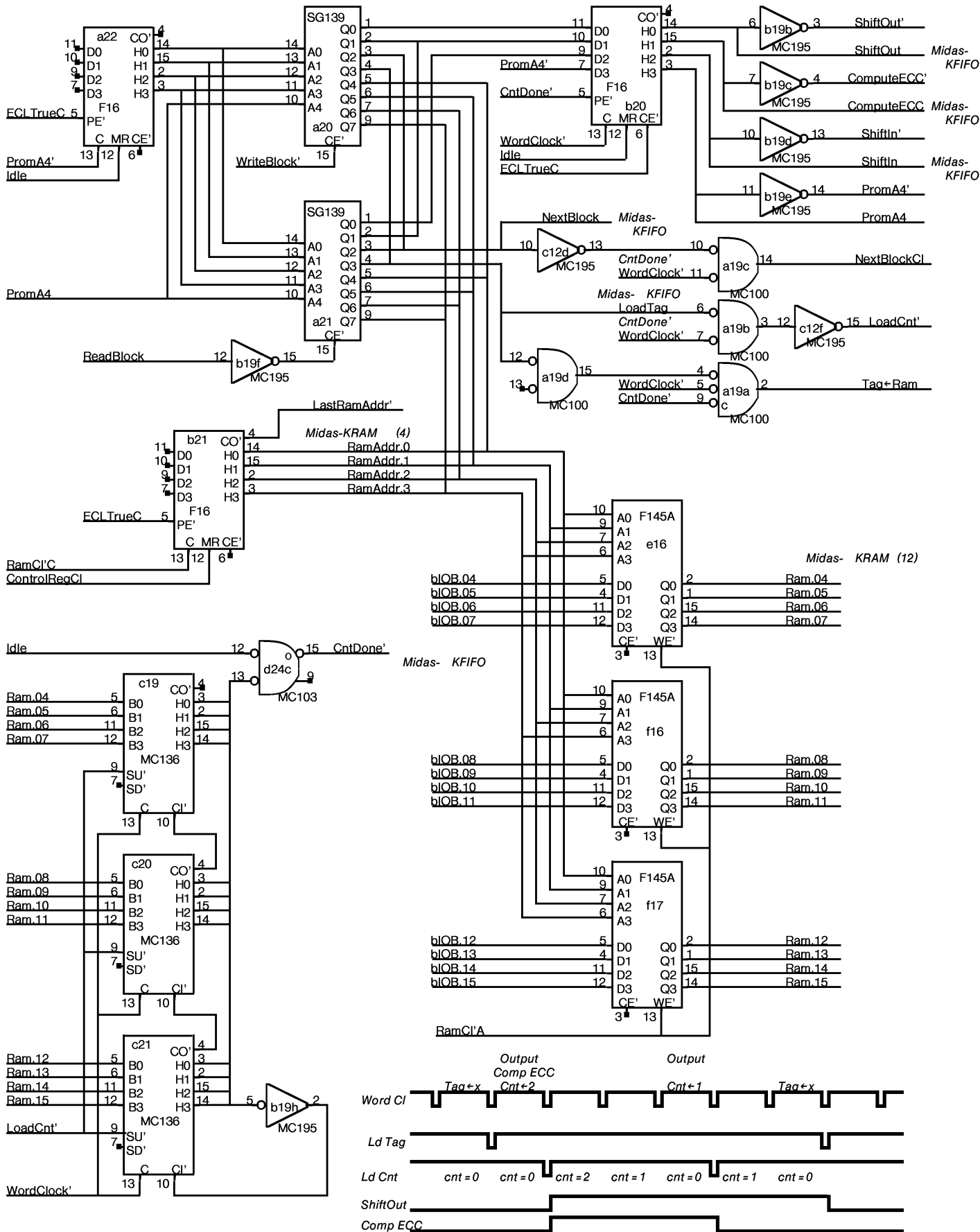
K1115A at i20



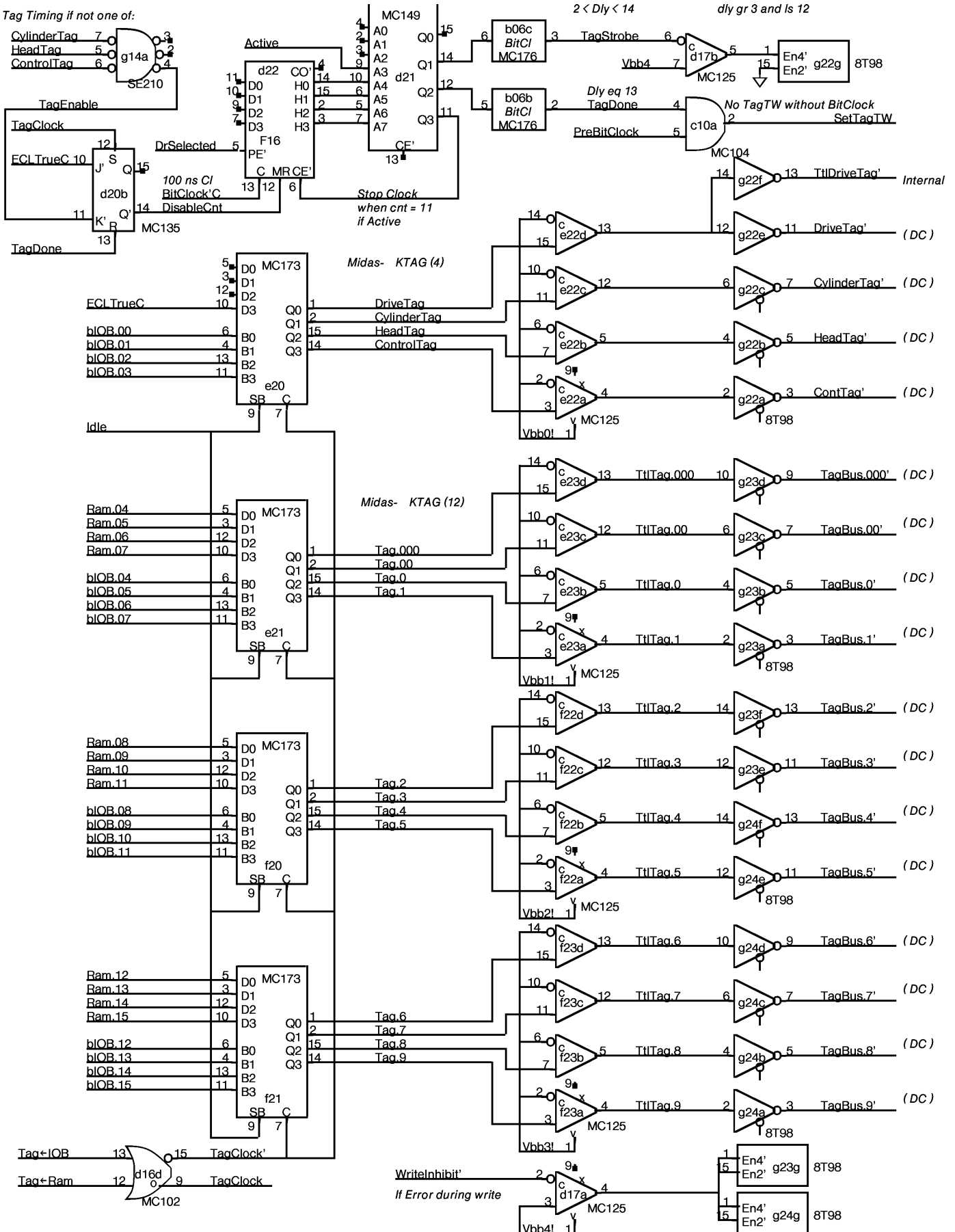
Logo is near pin 1.
This is a 14 pin pkg.
Board pins 8 & 9 aren't used.
Remove bypass cap above pin 1

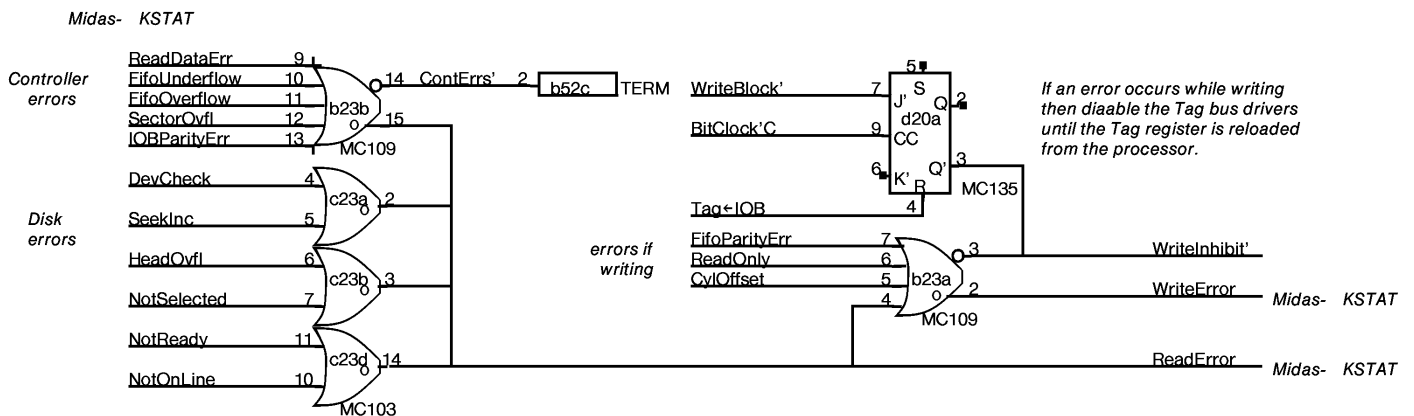
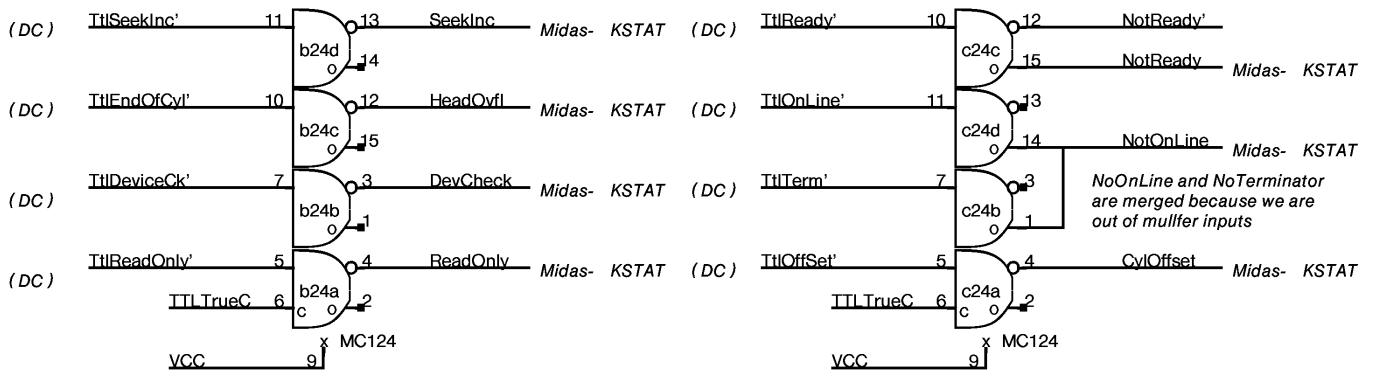
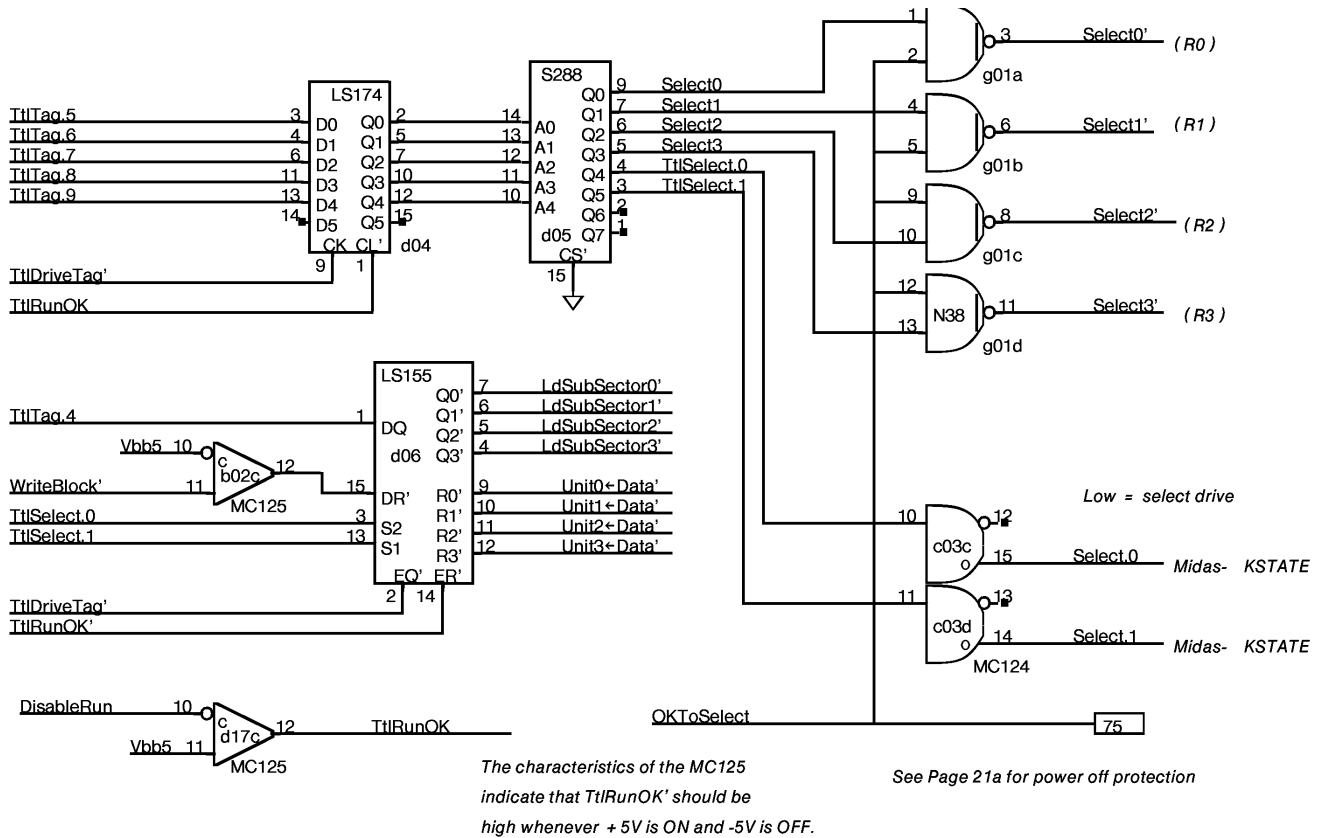


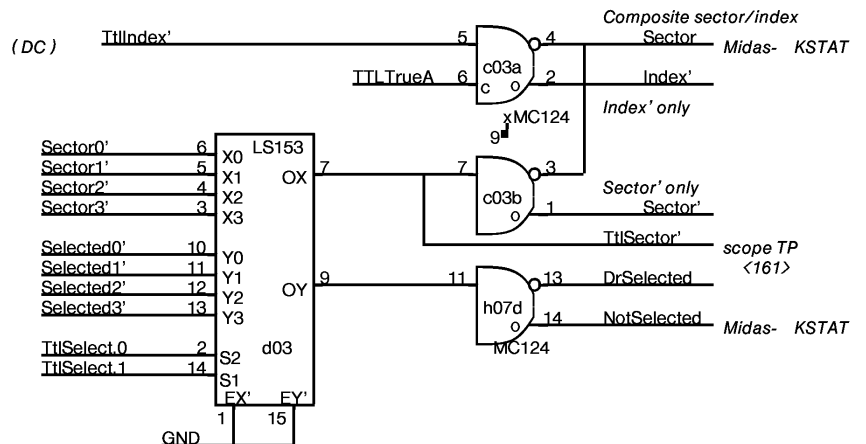
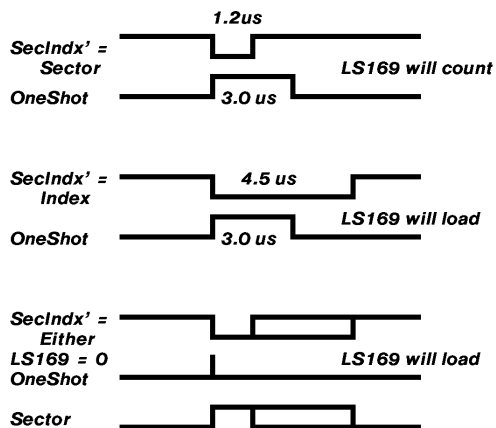
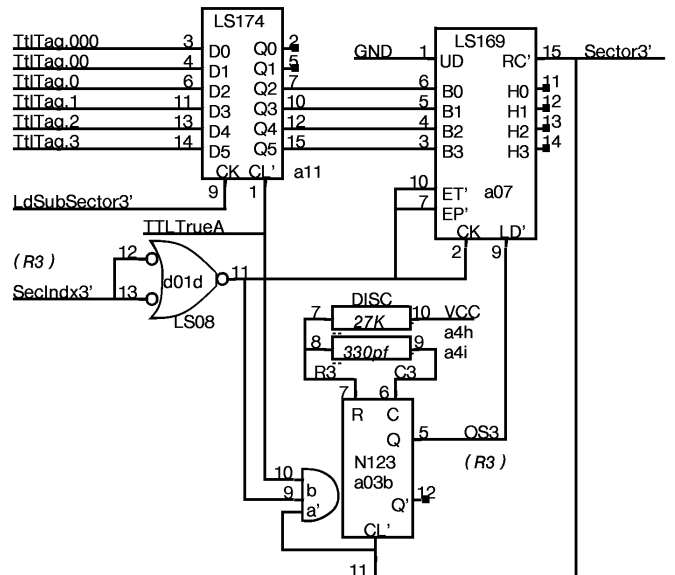
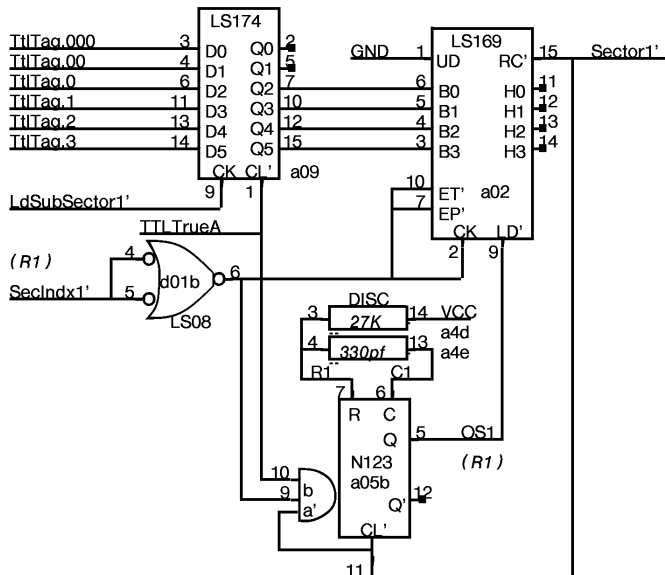
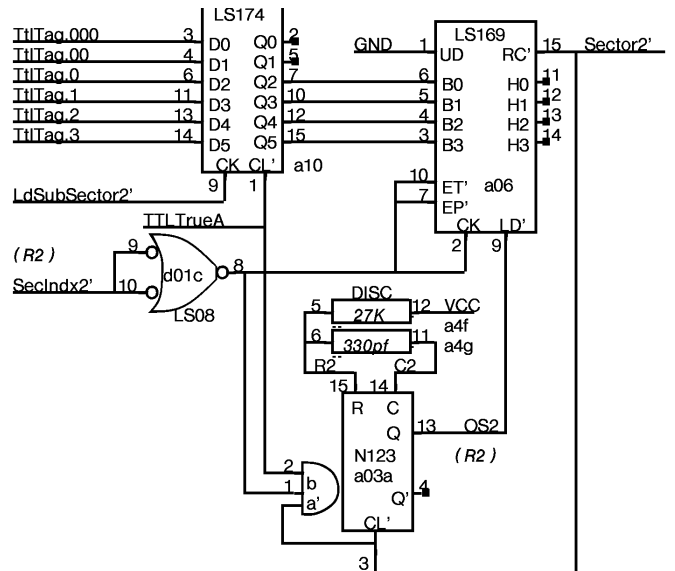
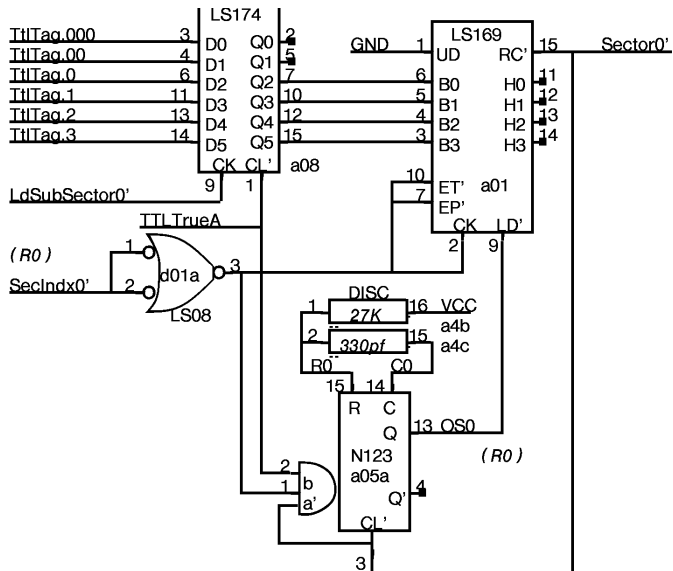




No Tag Timing if not one of:

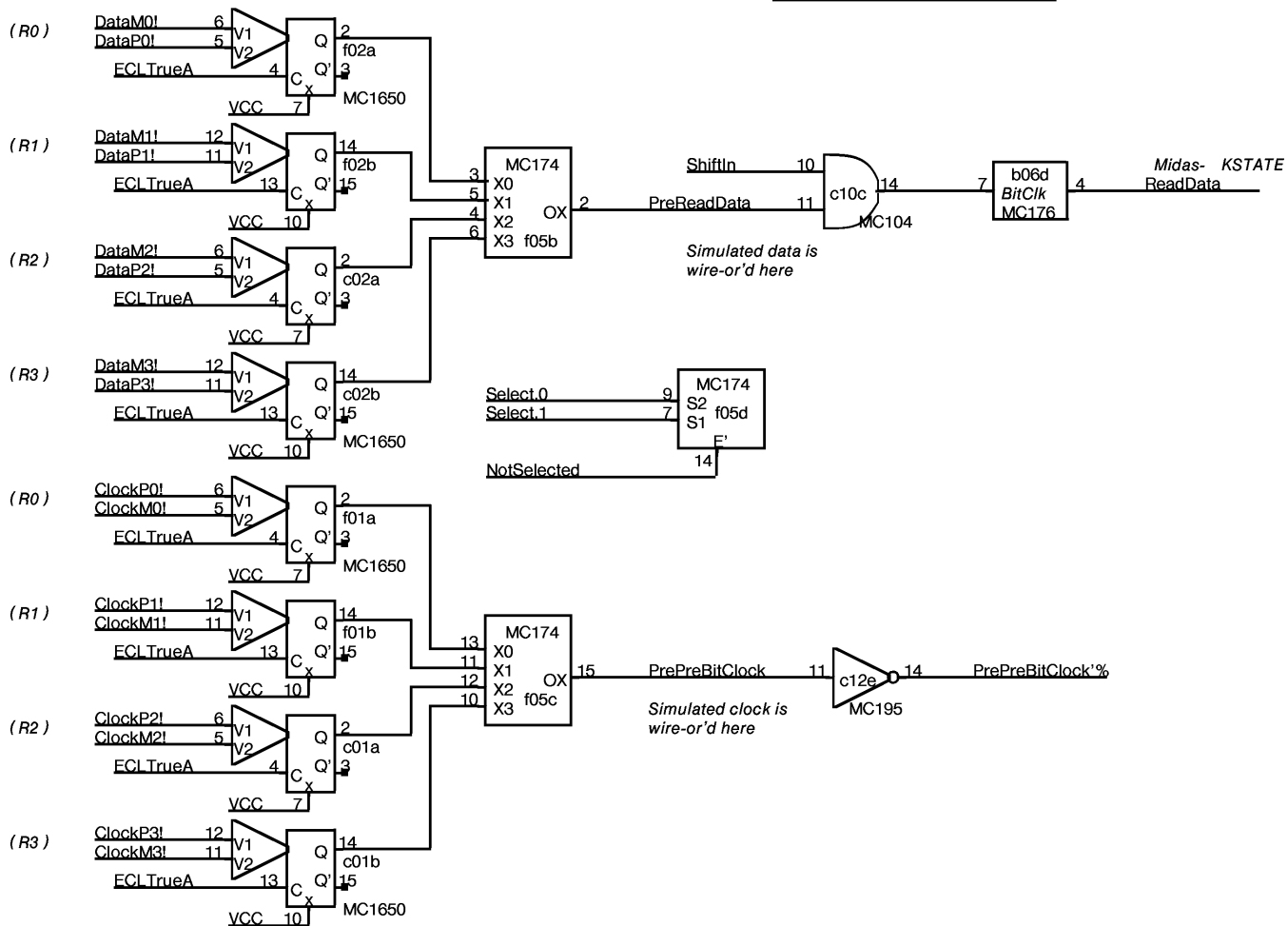
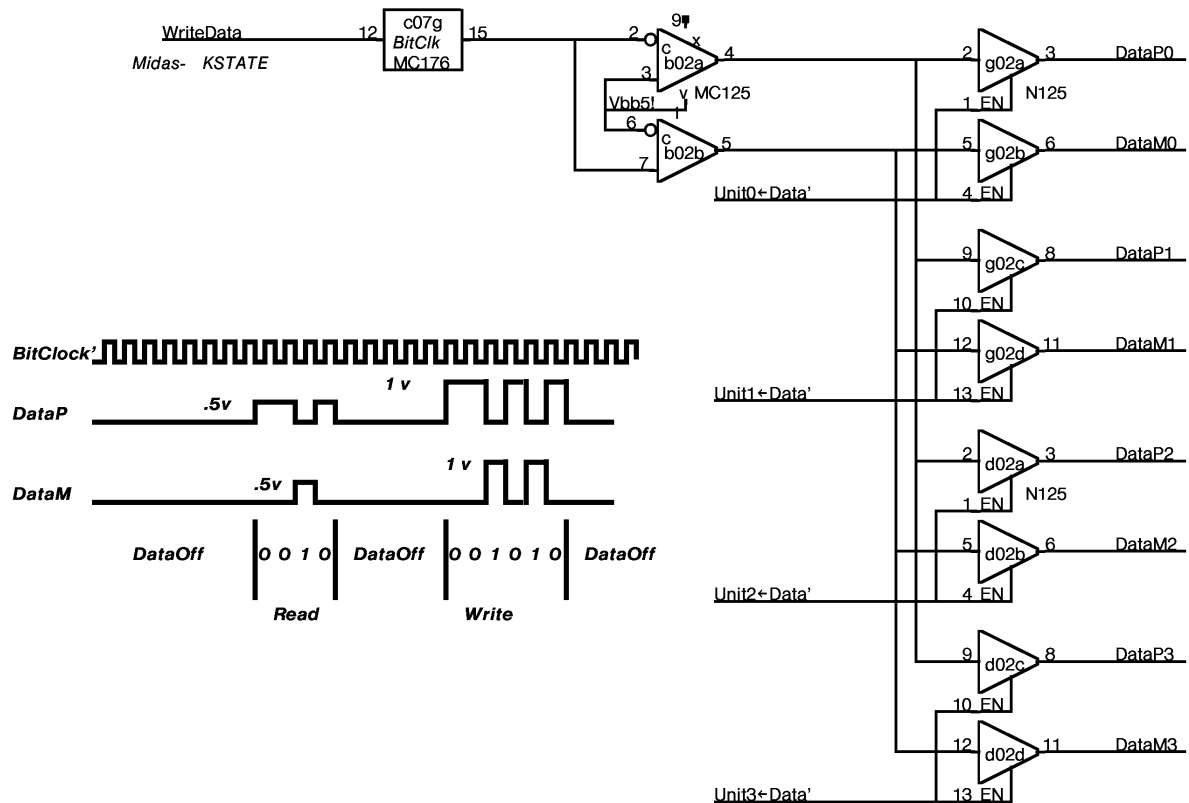


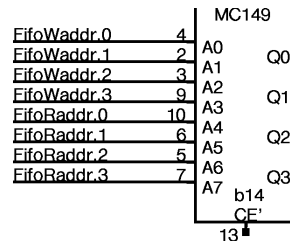
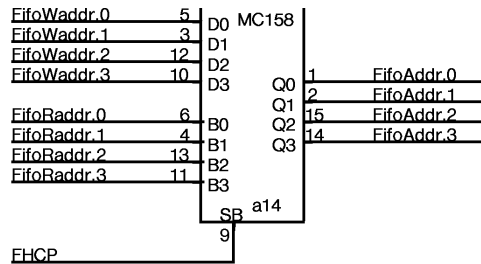




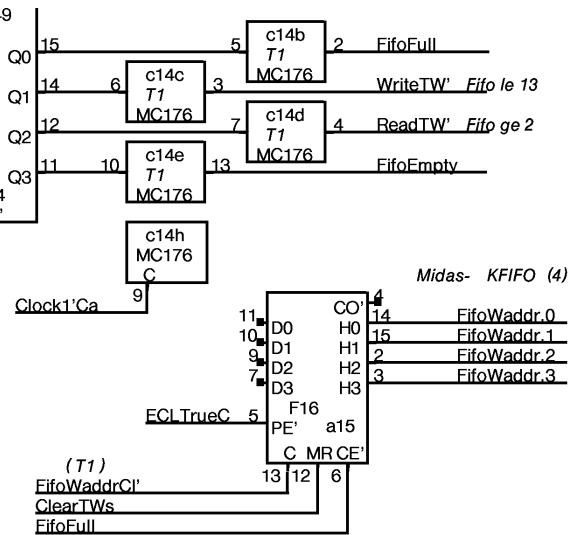
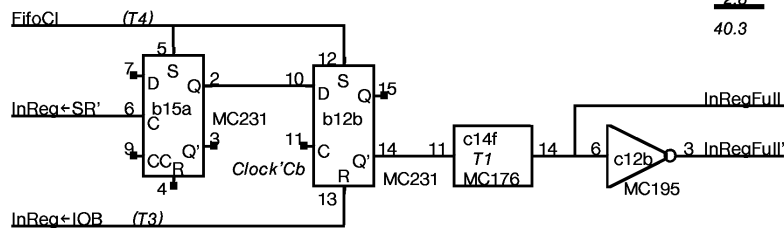
The above sector counters require the TRIDENT disk to have its sector counters set to provide 117 "sub-sector" pulses per revolution. This is done by setting the disk jumpers as follows:

X6A
4 - 11
5 - 10
X6B
2 - 13
3 - 12
4 - 11
5 - 10
6 - 09

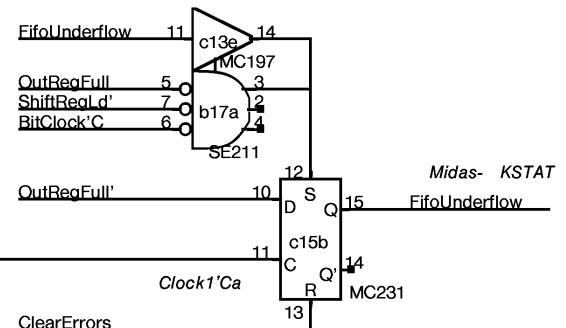
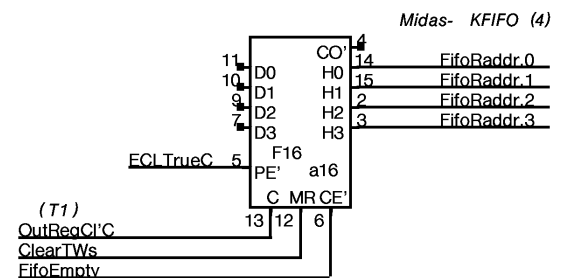
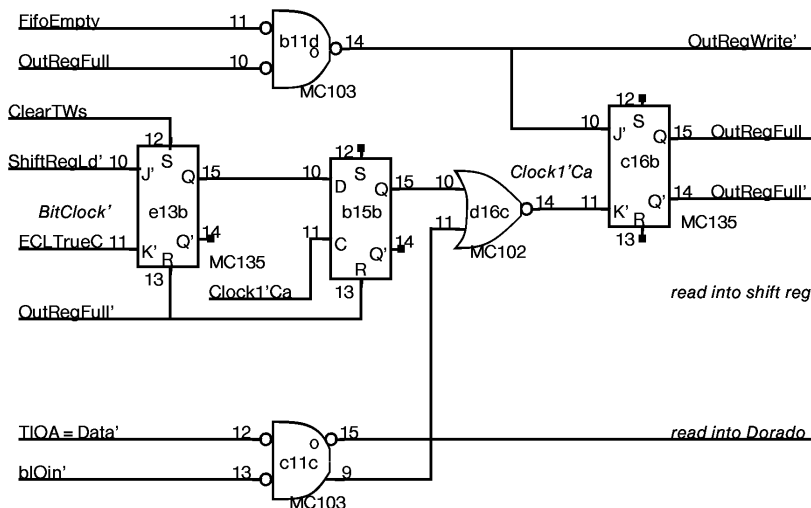
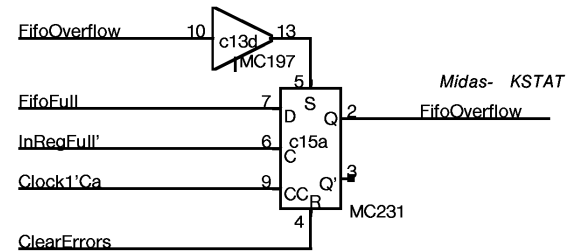
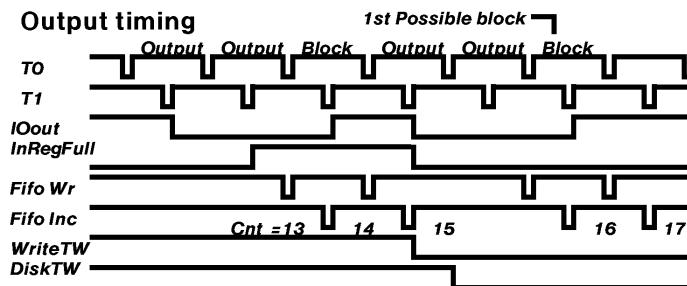




Delay Cnt = 5.5
 32
 2.8
 40.3

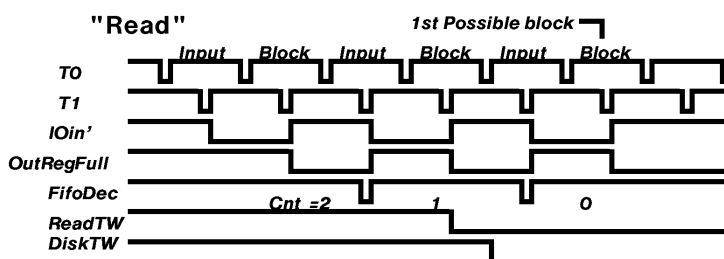


Output timing

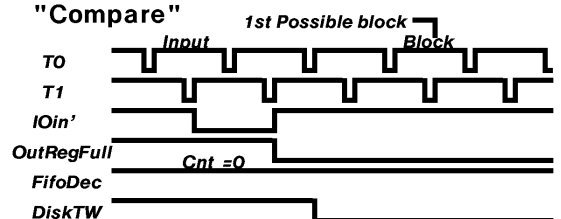


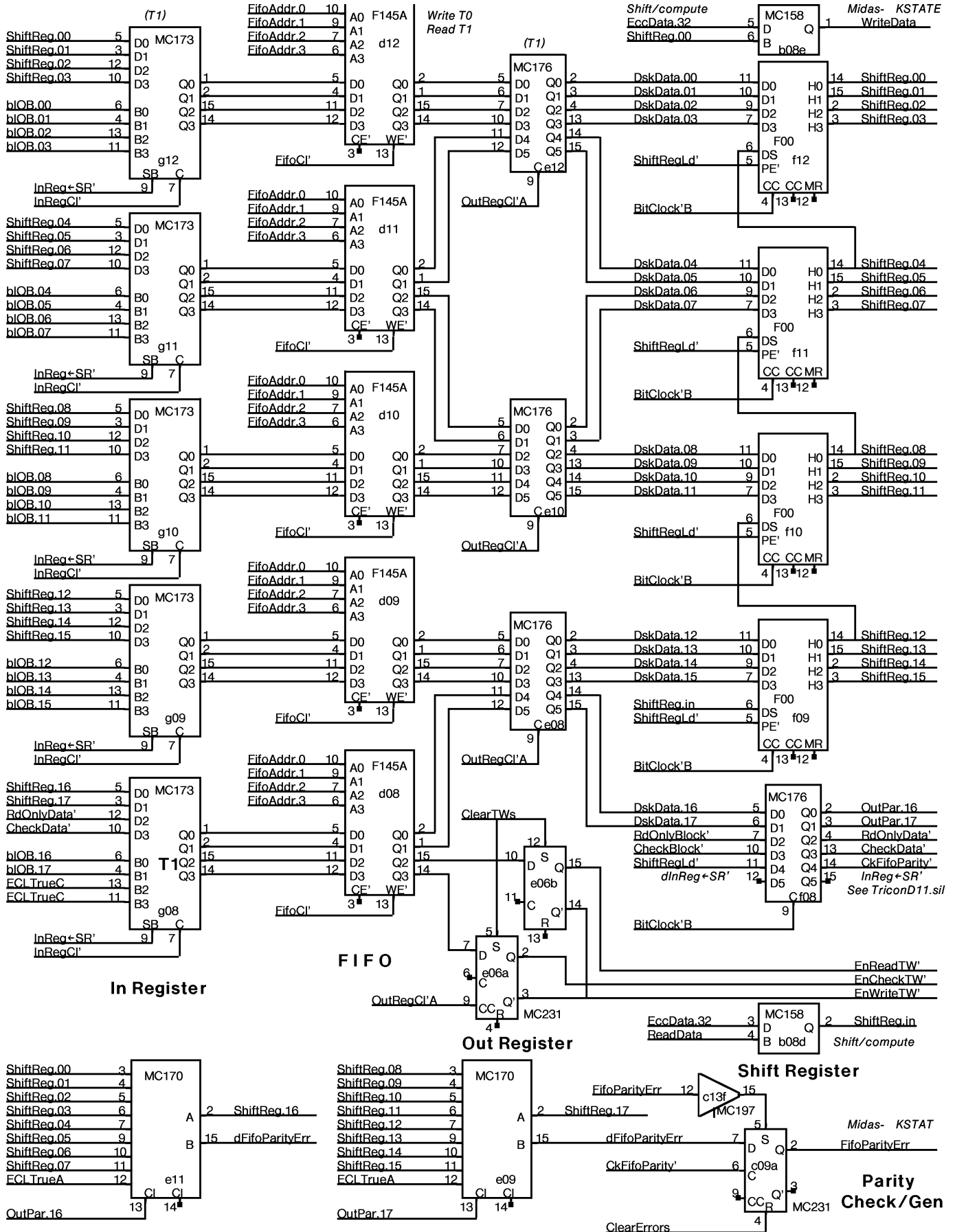
Input timing

"Read"

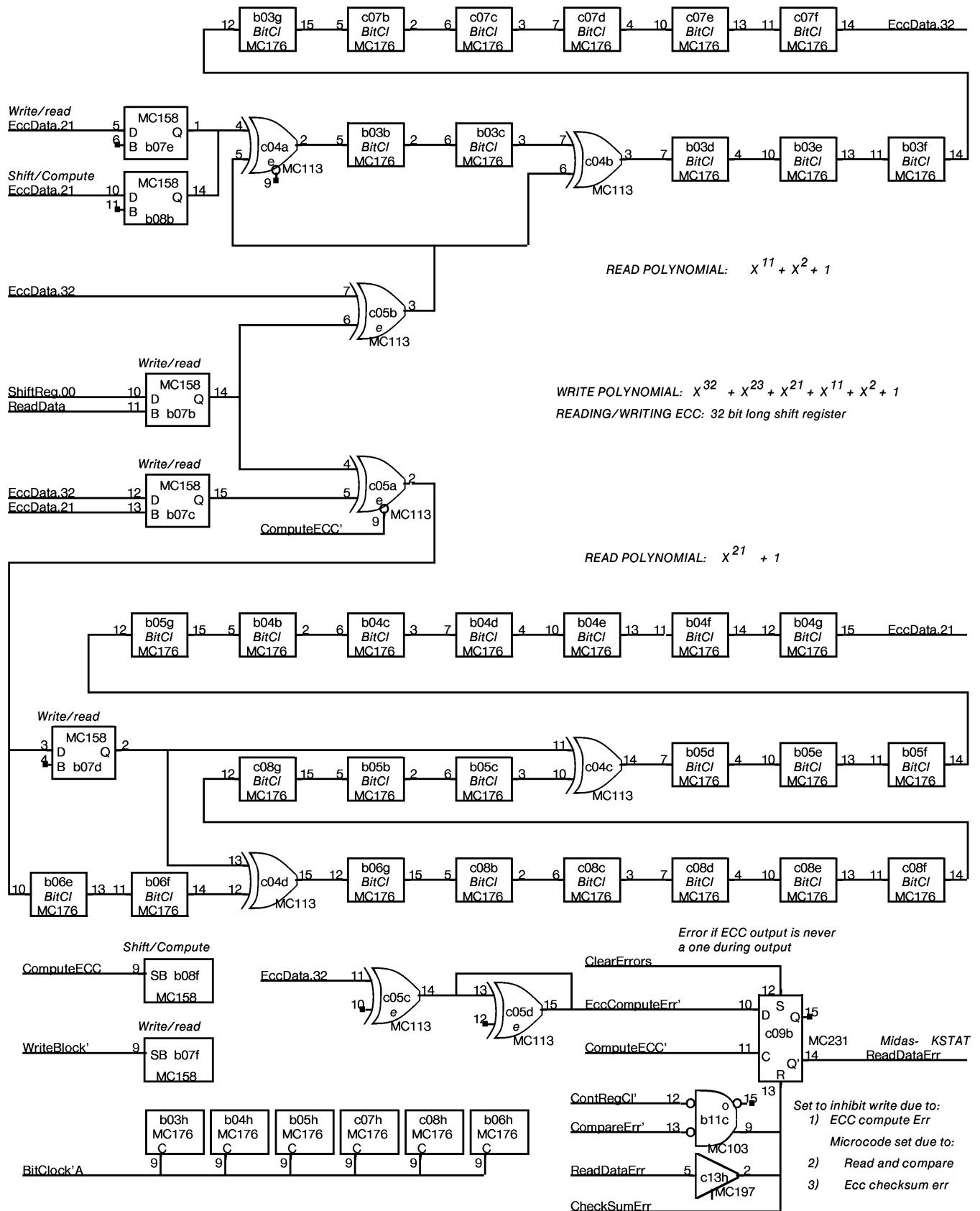


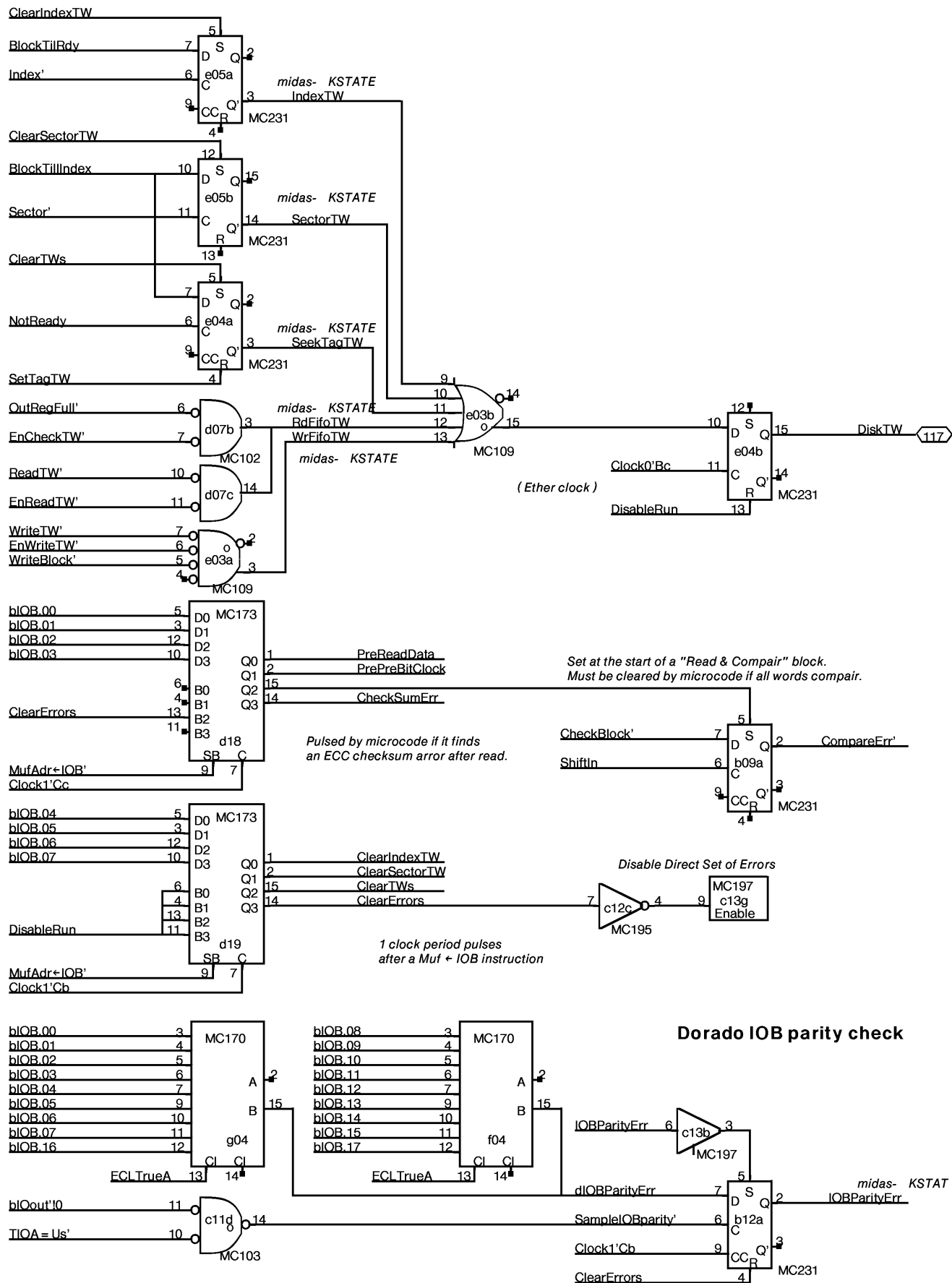
"Compare"



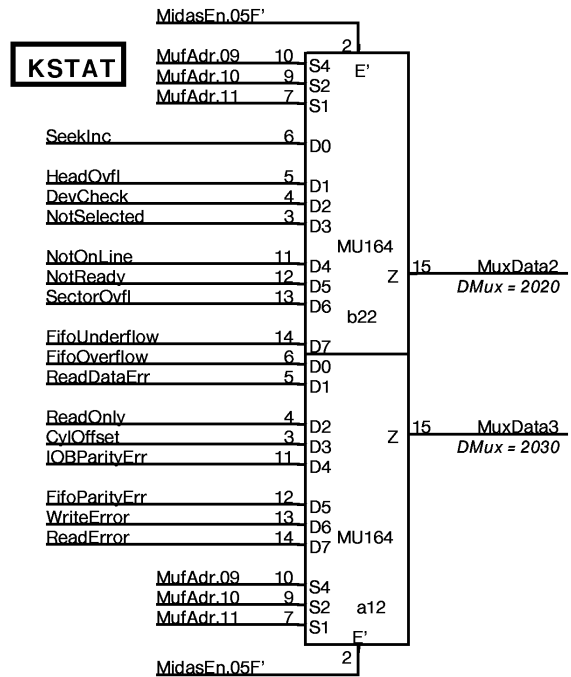
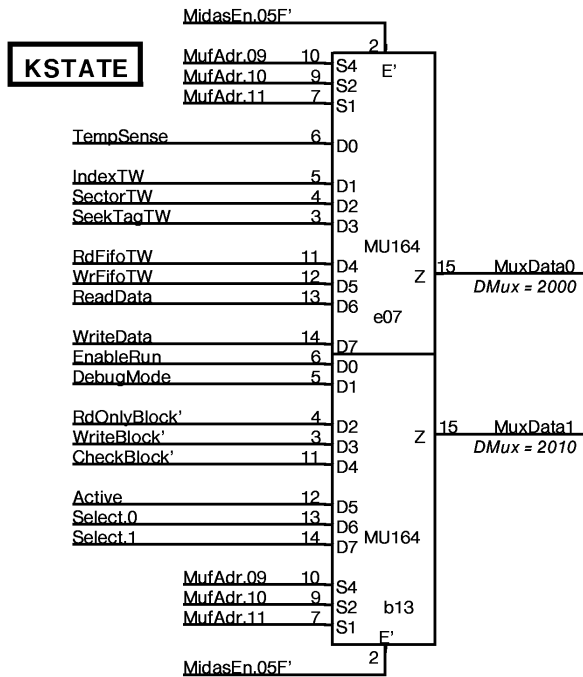


POLYNOMIAL DIVIDER FOR FIRE CODE GENERATION





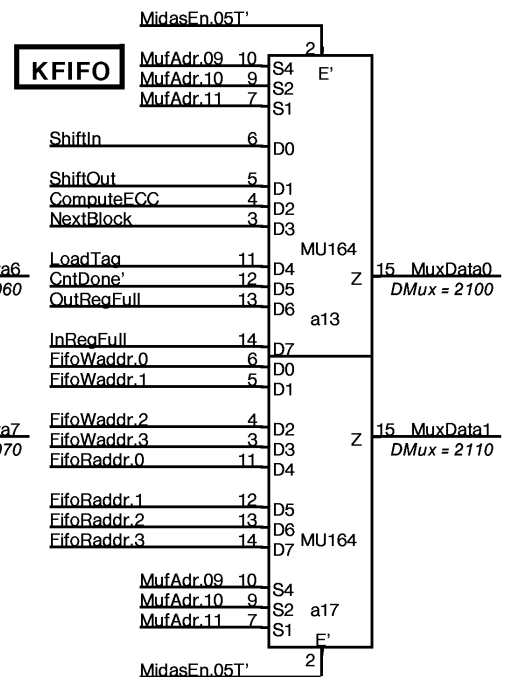
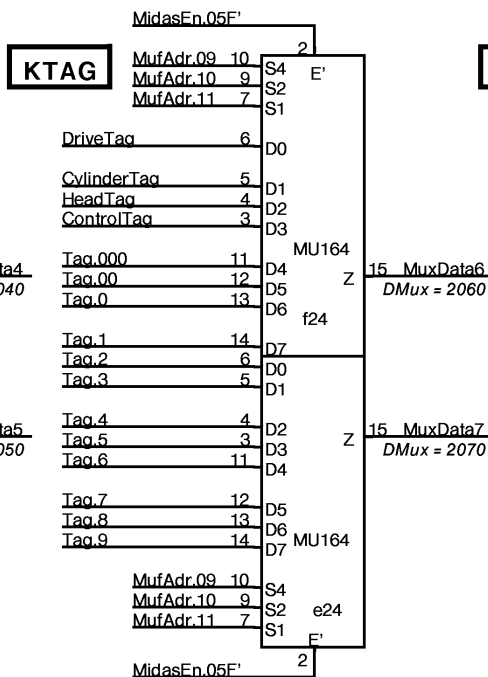
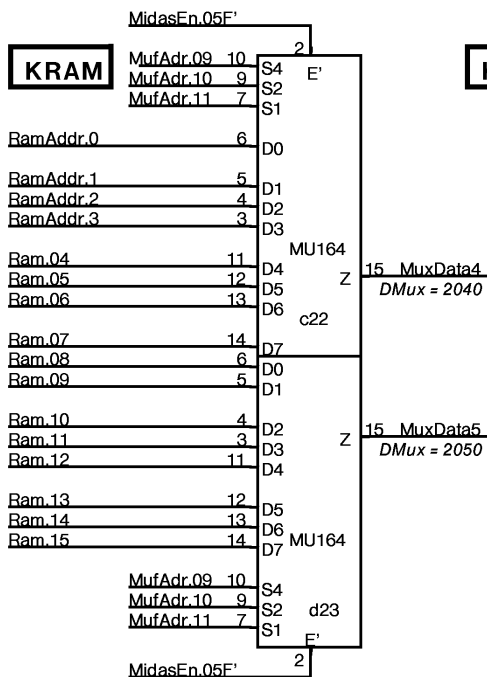
Dorado IOB parity check

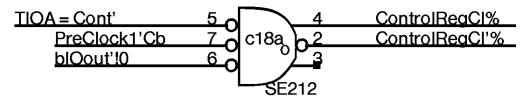
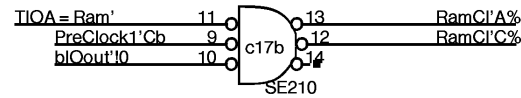
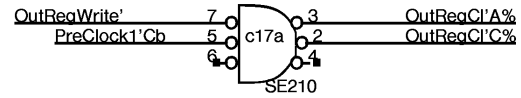
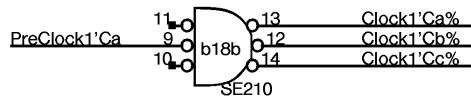
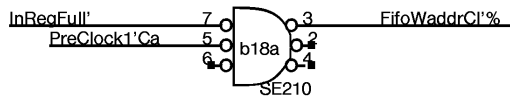
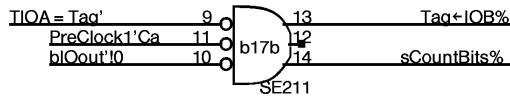
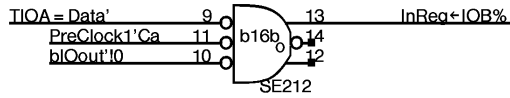
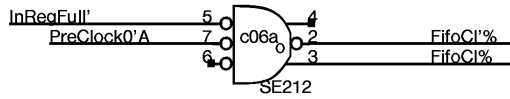
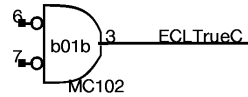
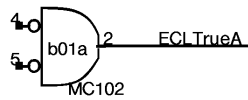


Muffler addresses are:

Value listed for Program input
Value plus 2000 for Midas input

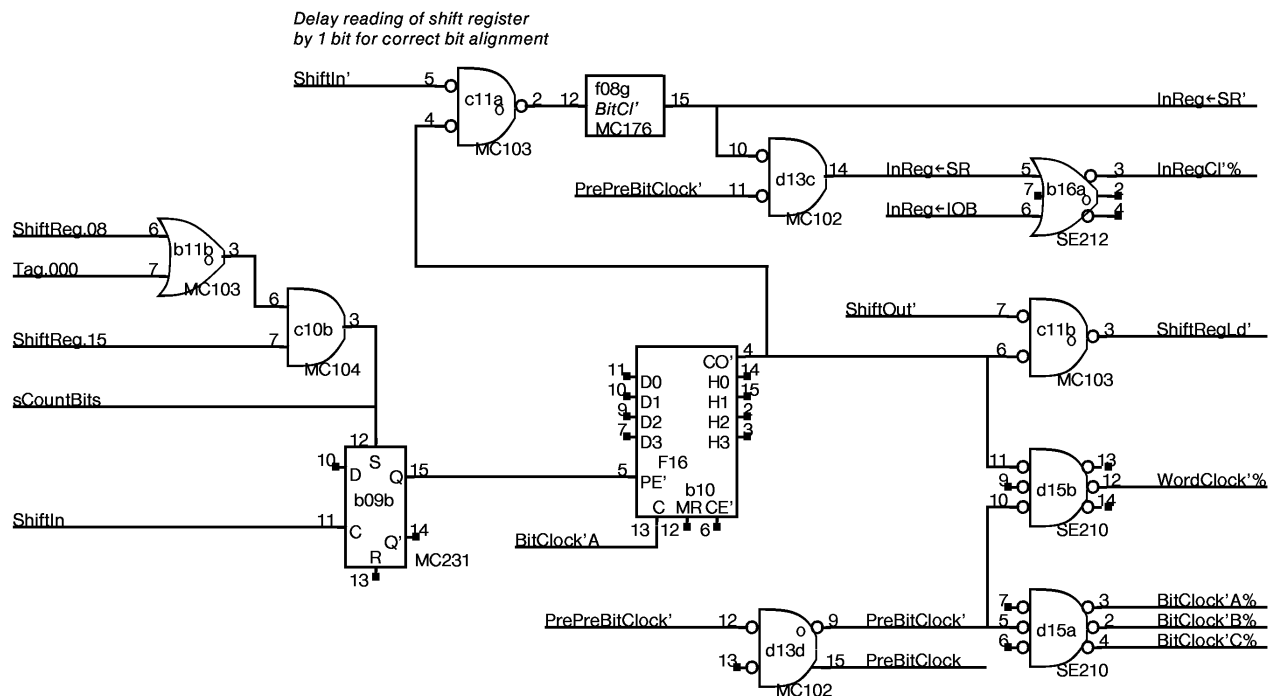
Values from 120 to 177 are
used by the Ethernet



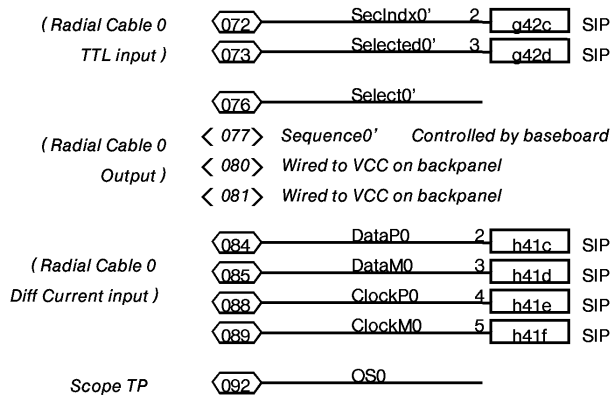


System Clocks

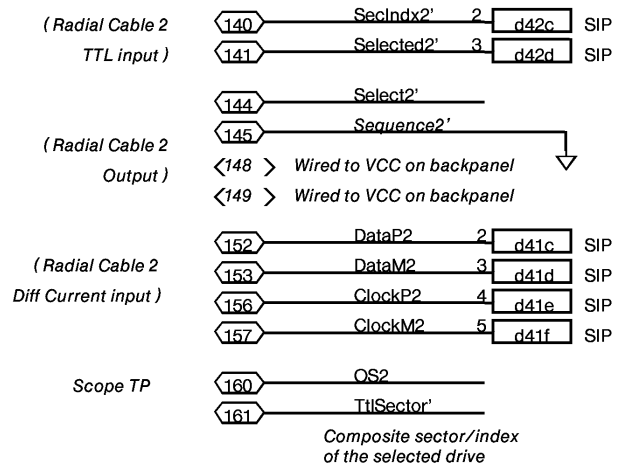
Disk Clocks



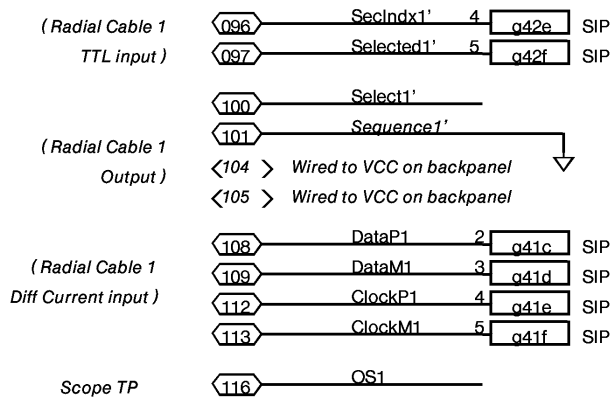
Radial Cable for Drive 0



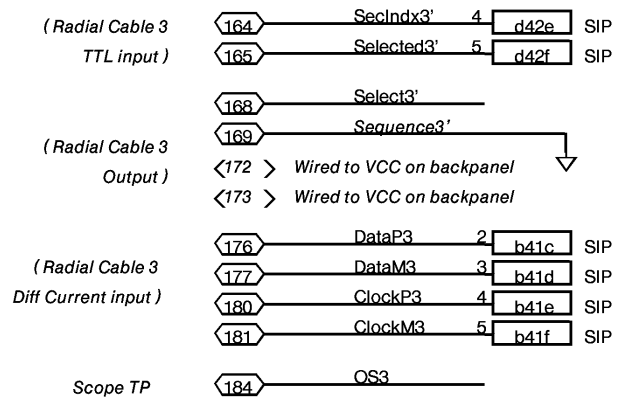
Radial Cable for Drive 2



Radial Cable for Drive 1

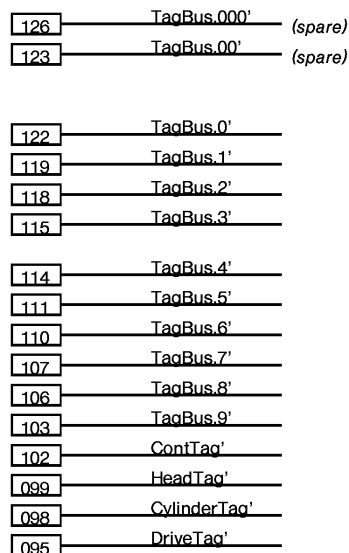


Radial Cable for Drive 3

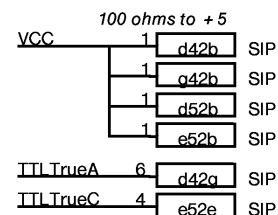
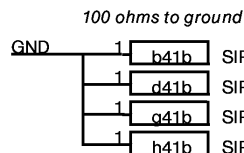
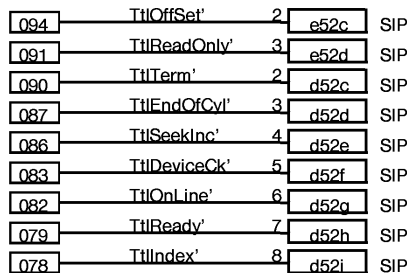


Daisy Chain Cable

Daisy chain outputs



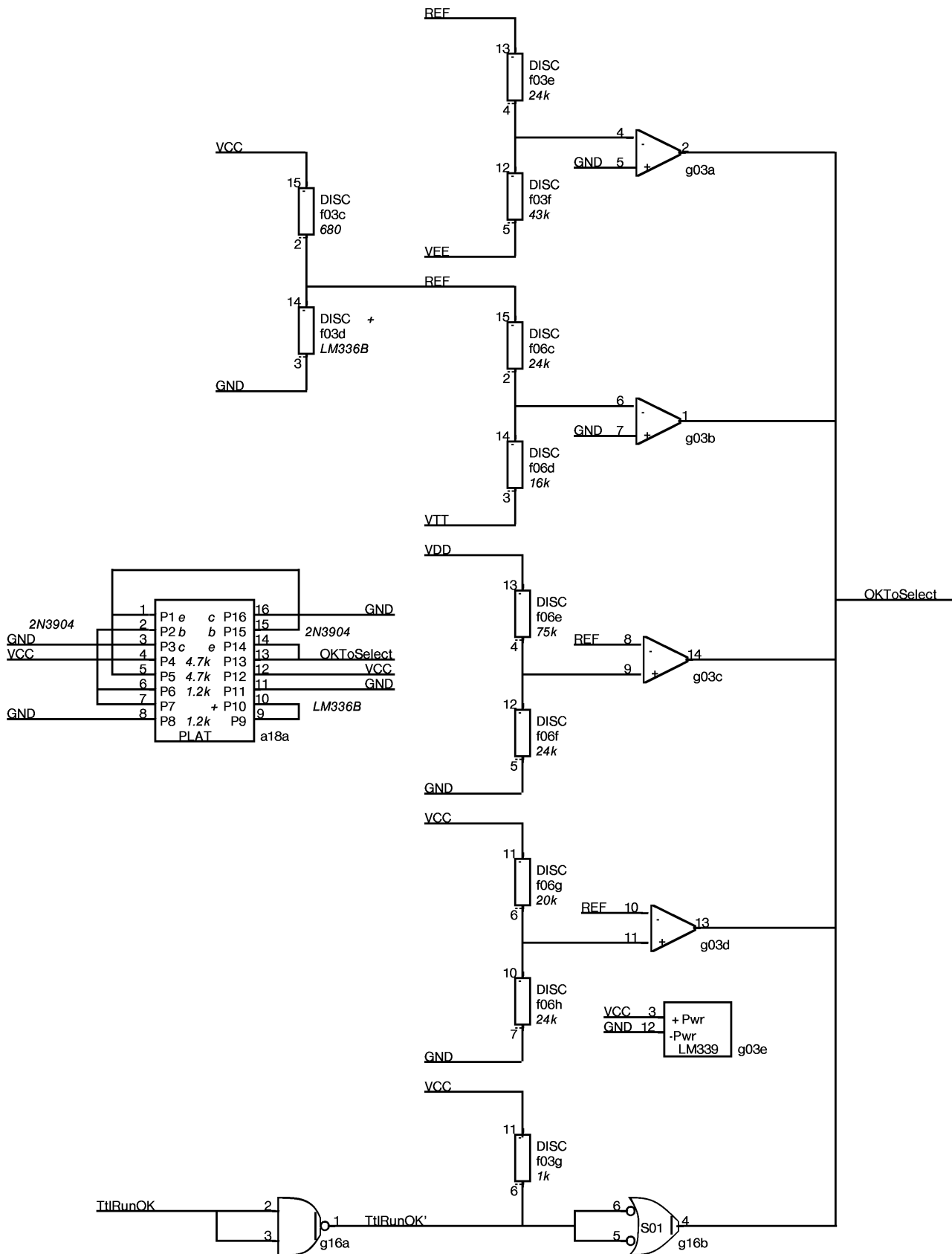
Daisy chain TTL inputs



Timing diagram for the 'be' signal. The signal is high for 8 clocks, then low for 24 clocks, and then high again. A note indicates the signal is 'turned off by ClearTW'.

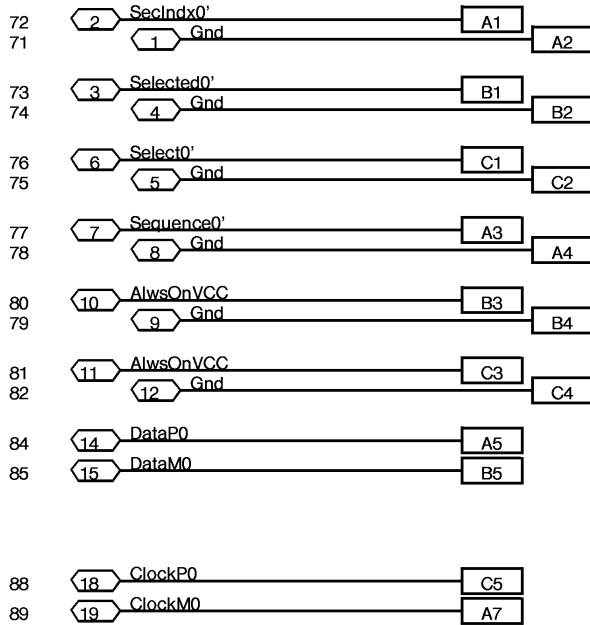
Timing diagram for the 1024-word data transfer sequence. The diagram shows the relationship between Sector/Index, ContTag' (102), TagBus.3' (f23.13), ReadData, and TagBus.2' (f23.11) signals. Key timing points include a 30 us delay after ContTag' assertion, a 25 us * delay between TagBus.3' and ReadData, and a <= 3 us delay between TagBus.2' and ReadData. The sequence involves writing a 2-word header, a 10-word label, and 1024-word data. The total time to end of data is 1.75 ms.

XEROX PARC	<i>Project</i> Dorado	<i>Reference</i> TriconD Timing Diagrams	<i>File</i> TriconD15.sil	<i>Designer</i> Roger Bates	<i>Rev</i> Cf	<i>Date</i> 7/08/79	<i>Page</i> 21
-----------------------------	--------------------------	---	------------------------------	--------------------------------	------------------	------------------------	-------------------



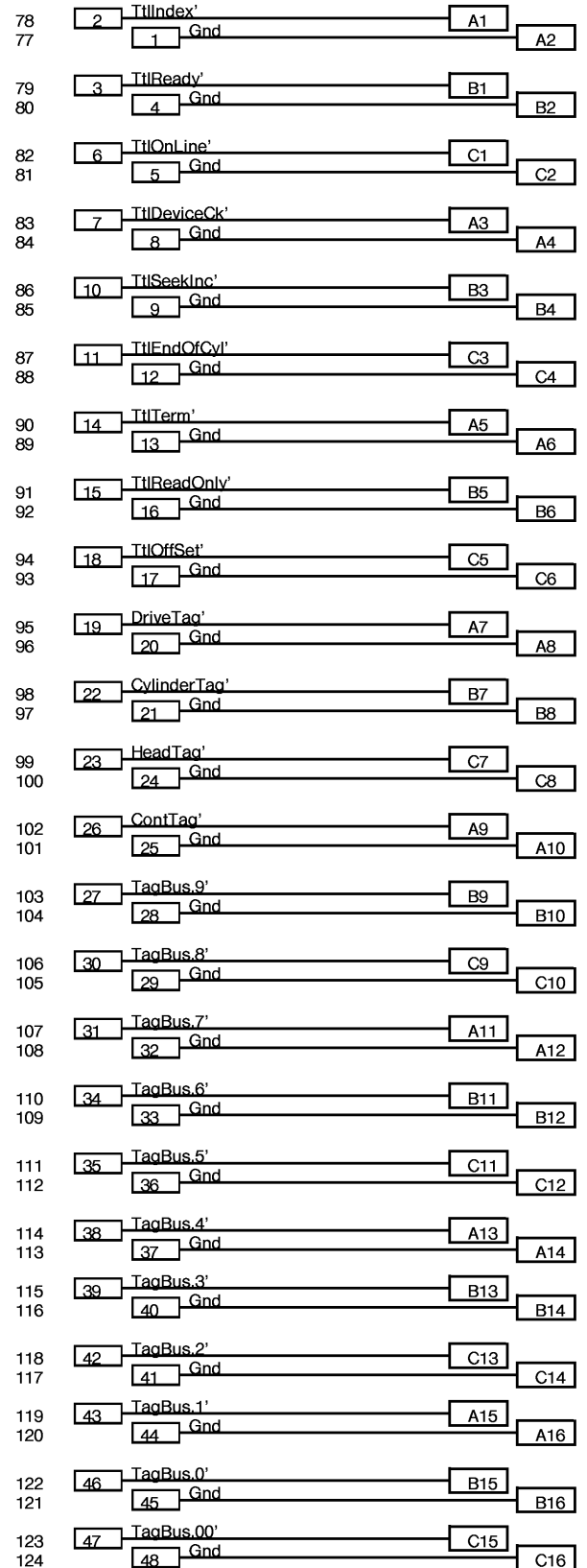
Radial Cable for Drive 0

AMP 204729-1



DAISEY CHAIN CABLE

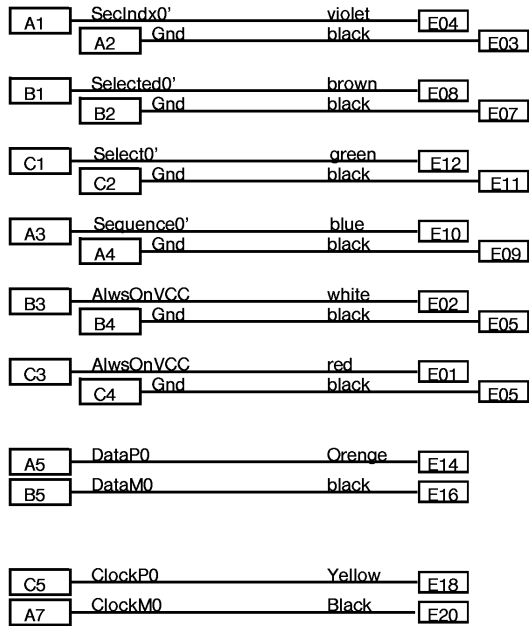
AMP 204733-1



Radial Cable for Drive 0

AMP 204742-1

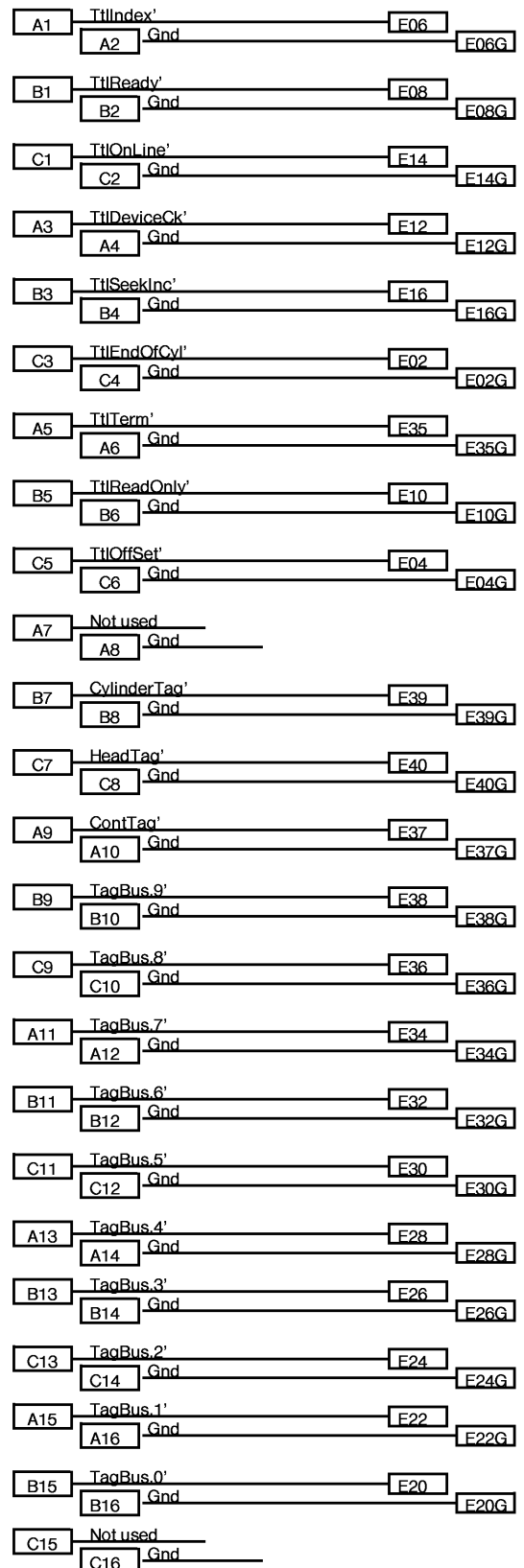
Cal-Comp
Assembly 12433

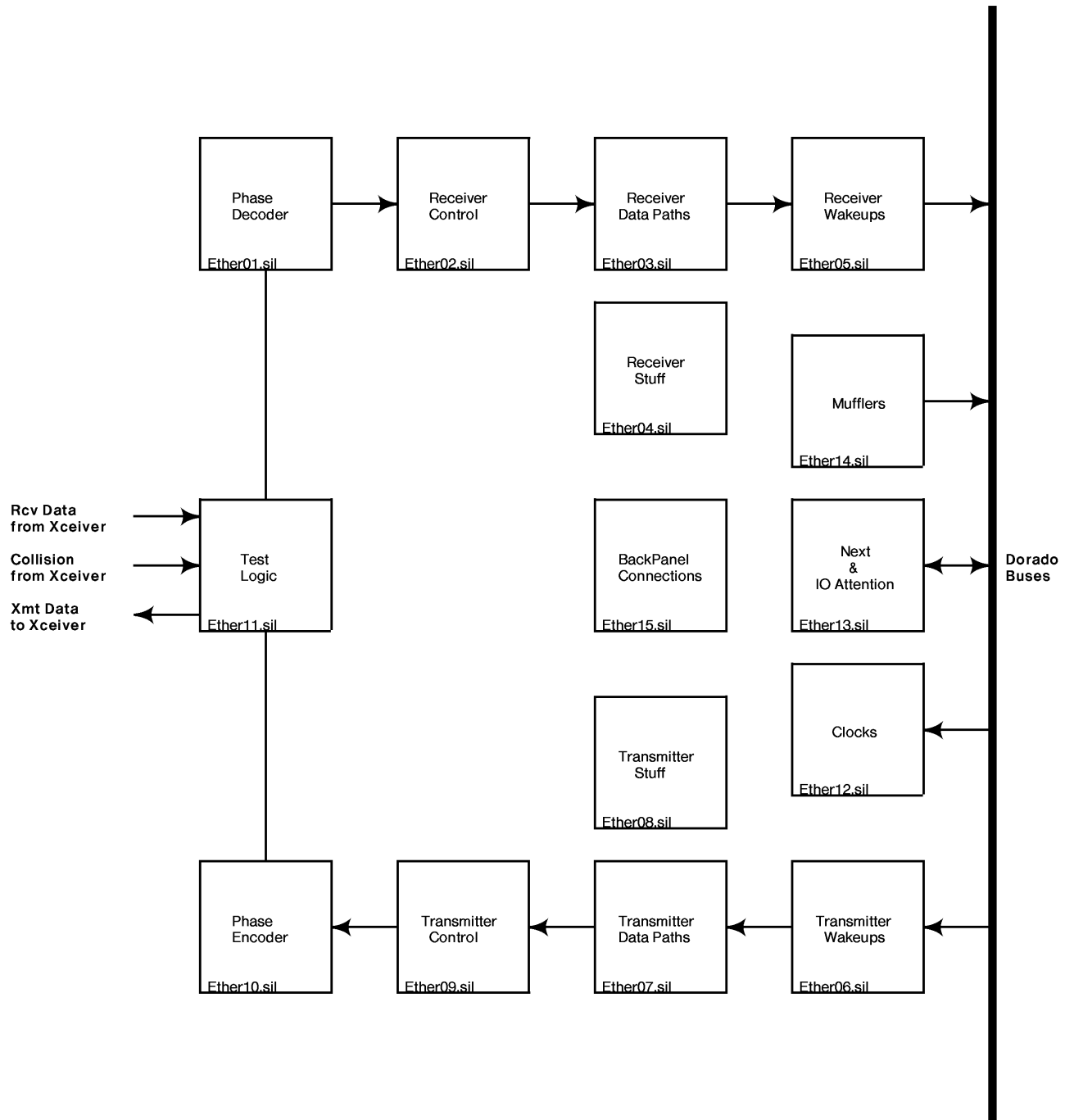


DAISEY CHAIN CABLE

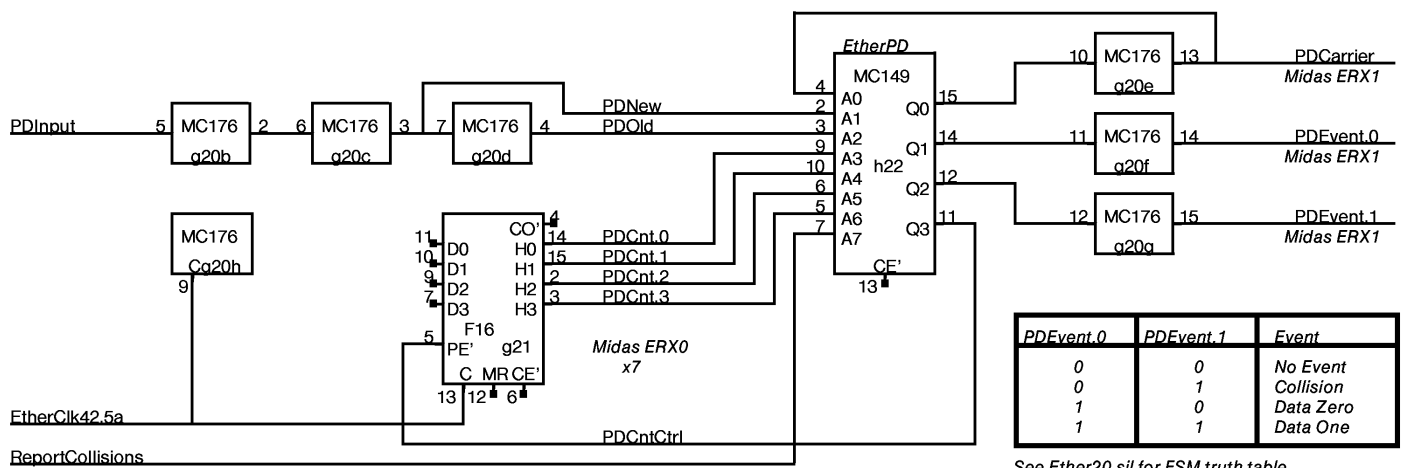
AMP 204746-1

Cal-Comp
Assembly 12424

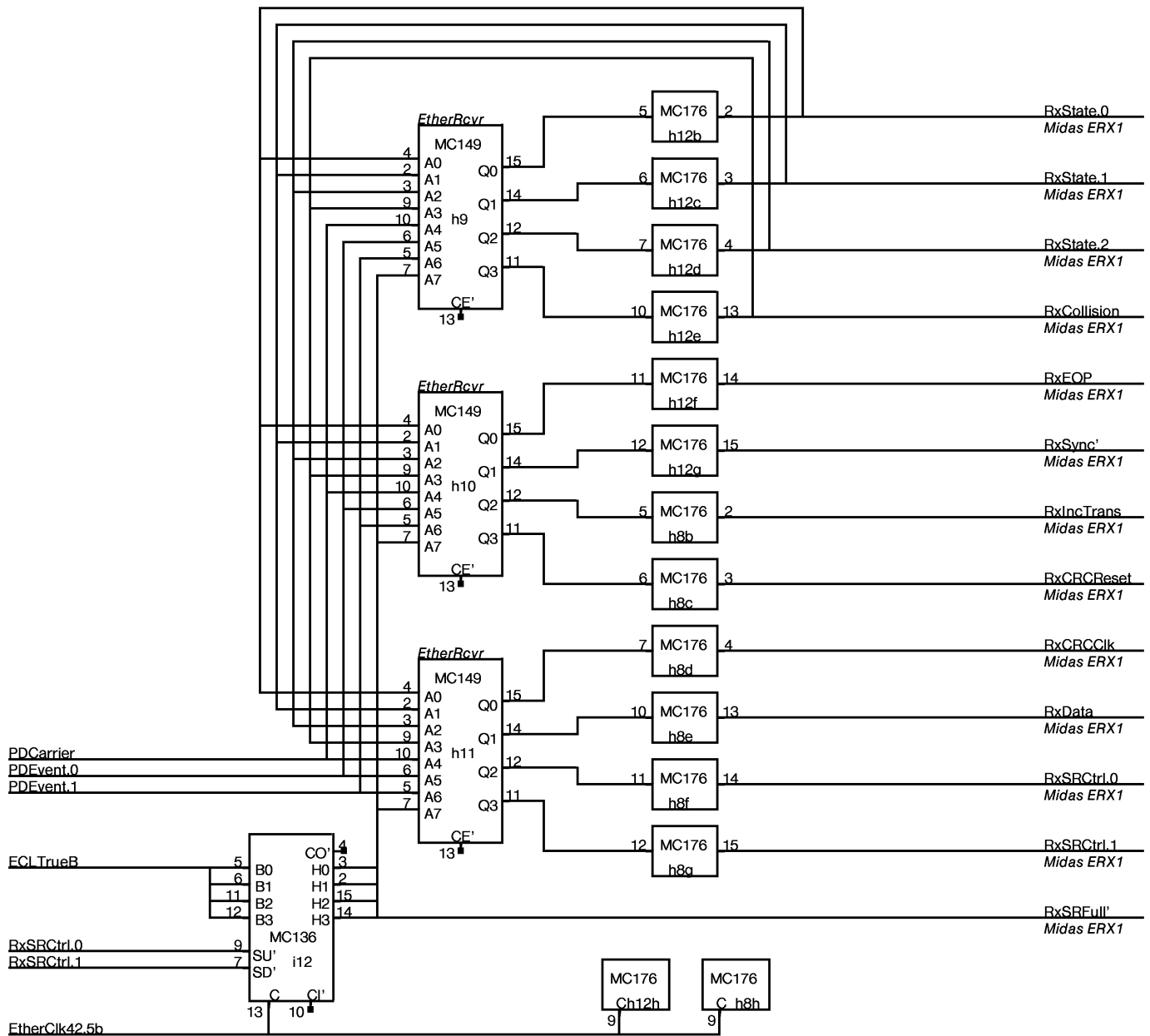




See DskEth*.sil for IOA, IOB, Muffler Control and Board Clocks

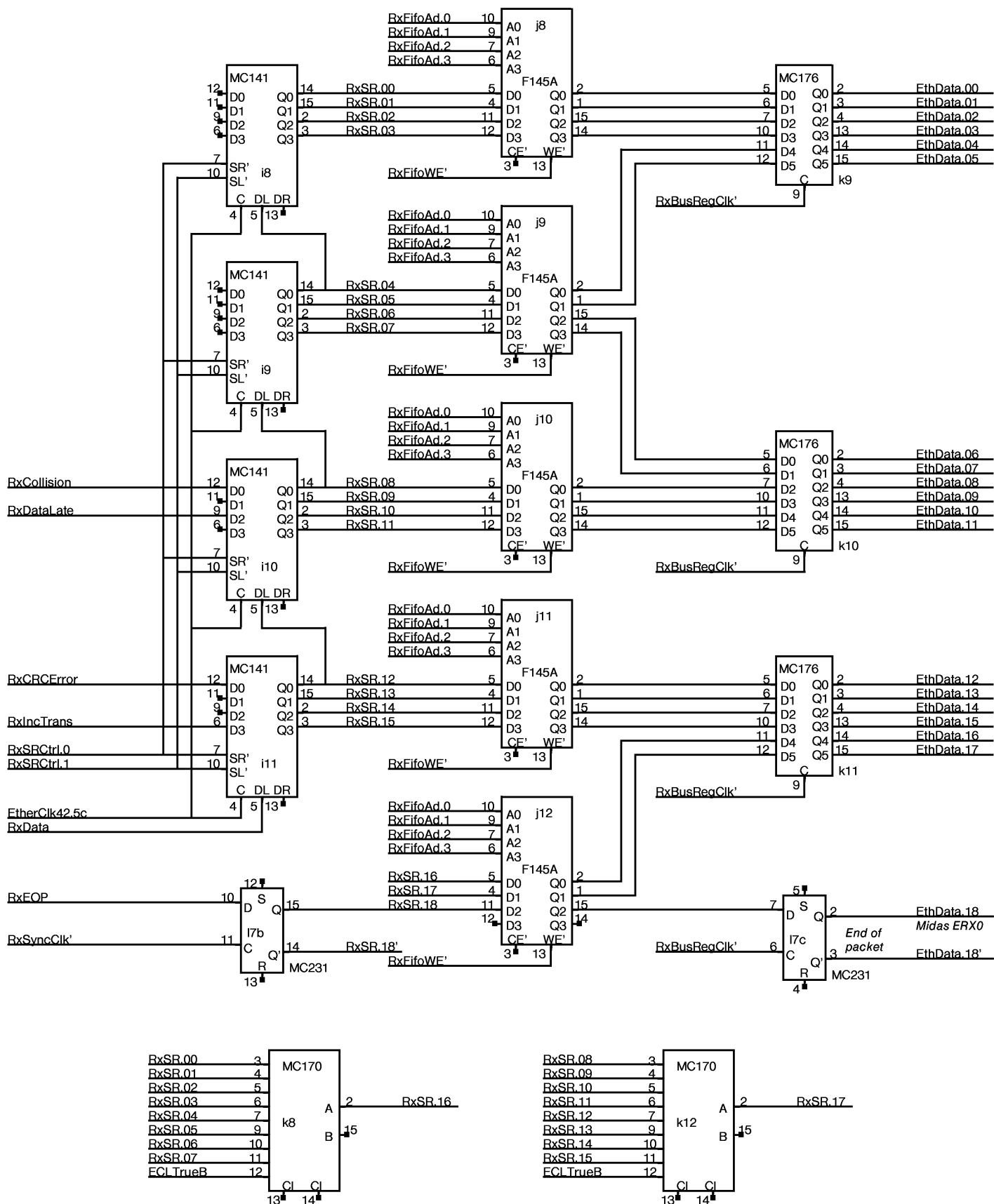


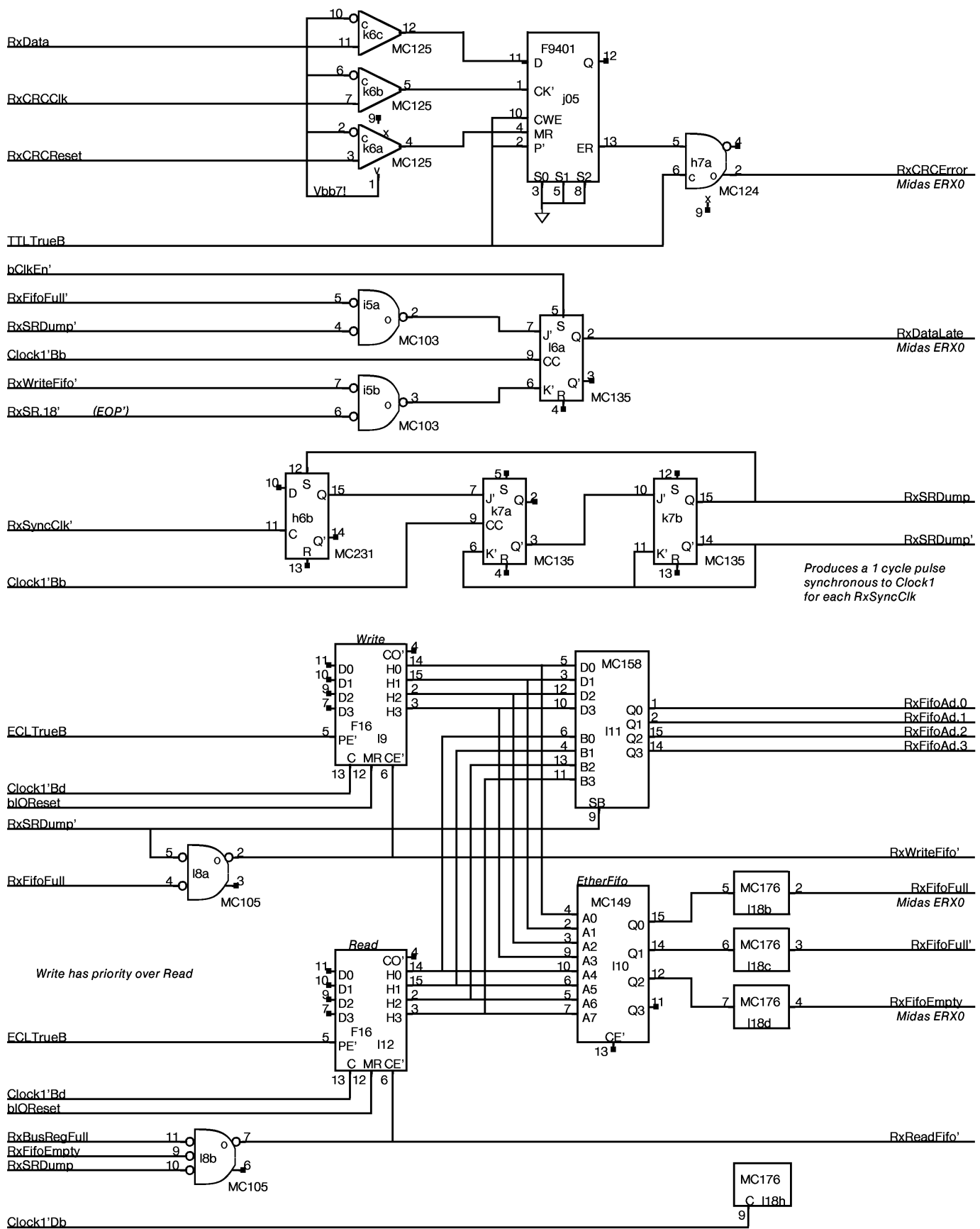
See Ether20.sil for FSM truth table

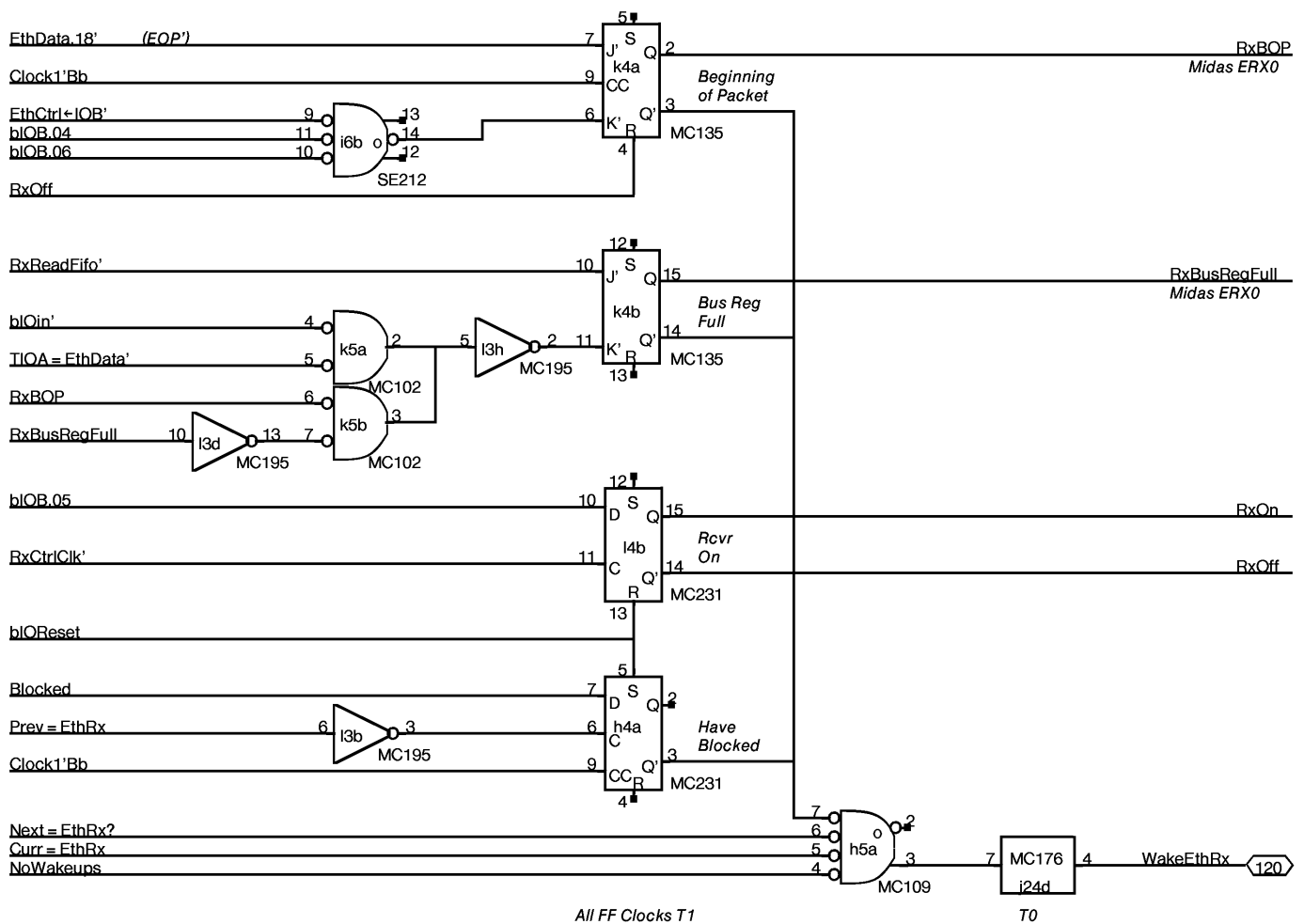


See Ether18.sil for timing diagrams

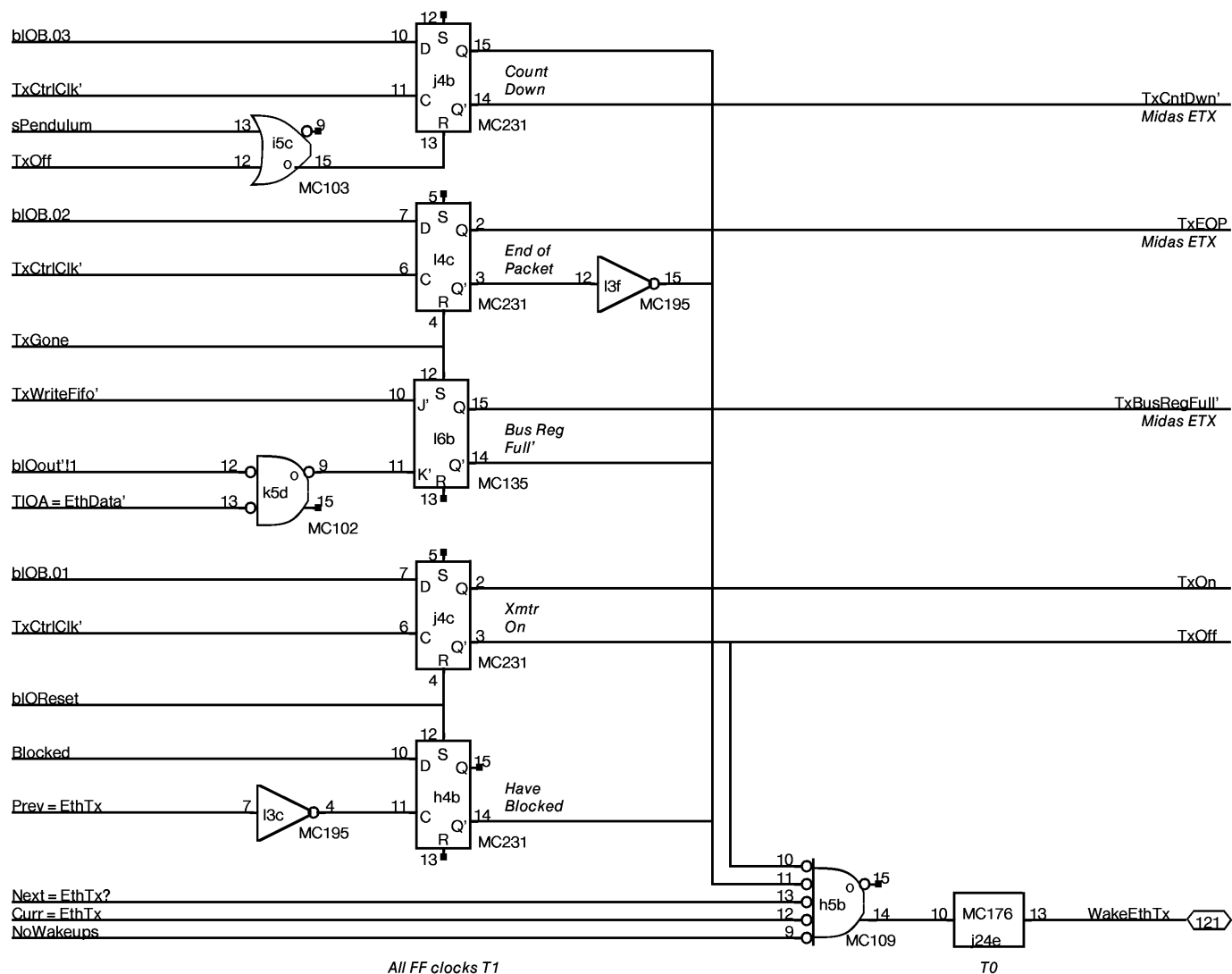
The slowest Dorado clock speed at which the receiver works is 85 ns (T0 to T1)



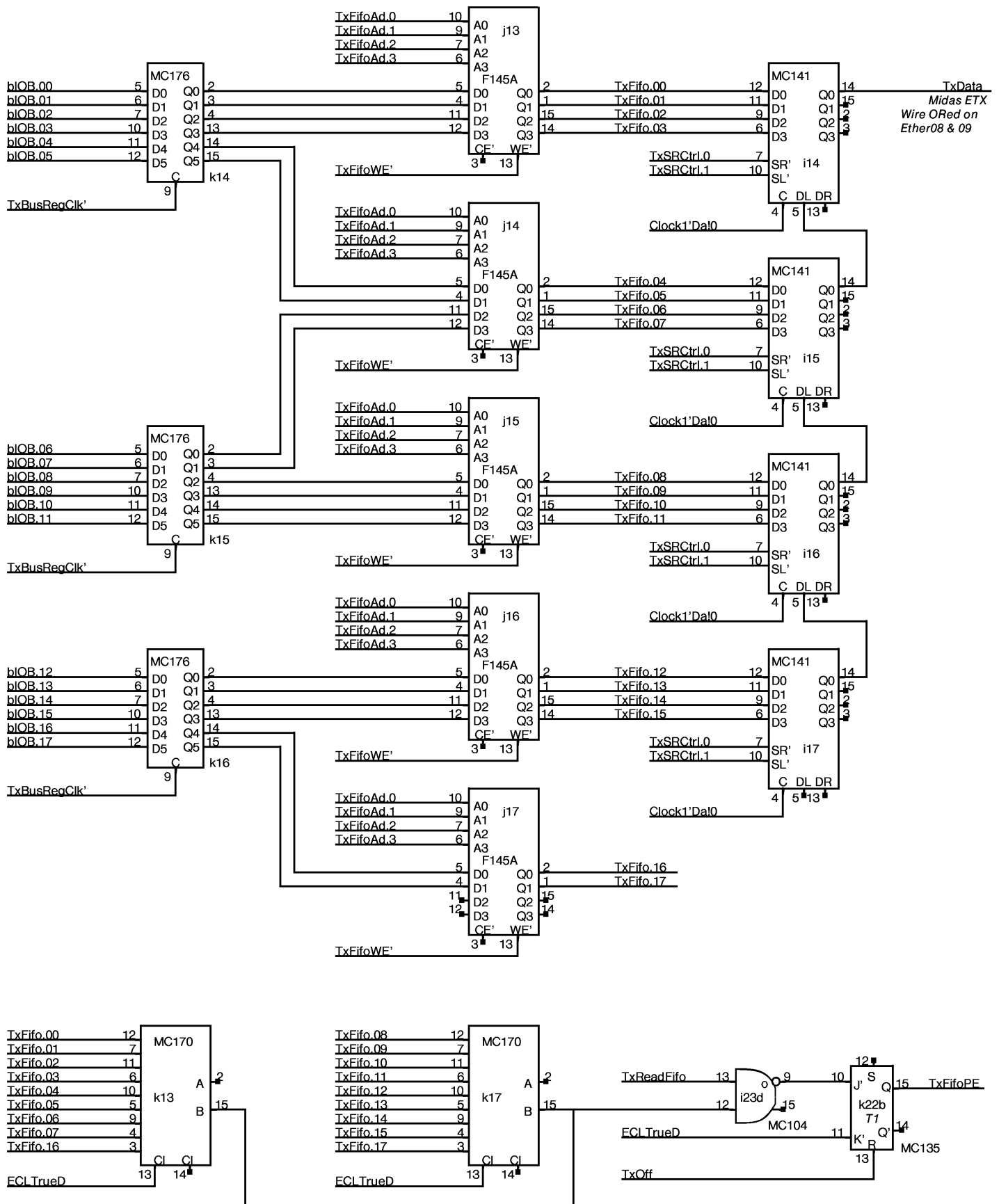


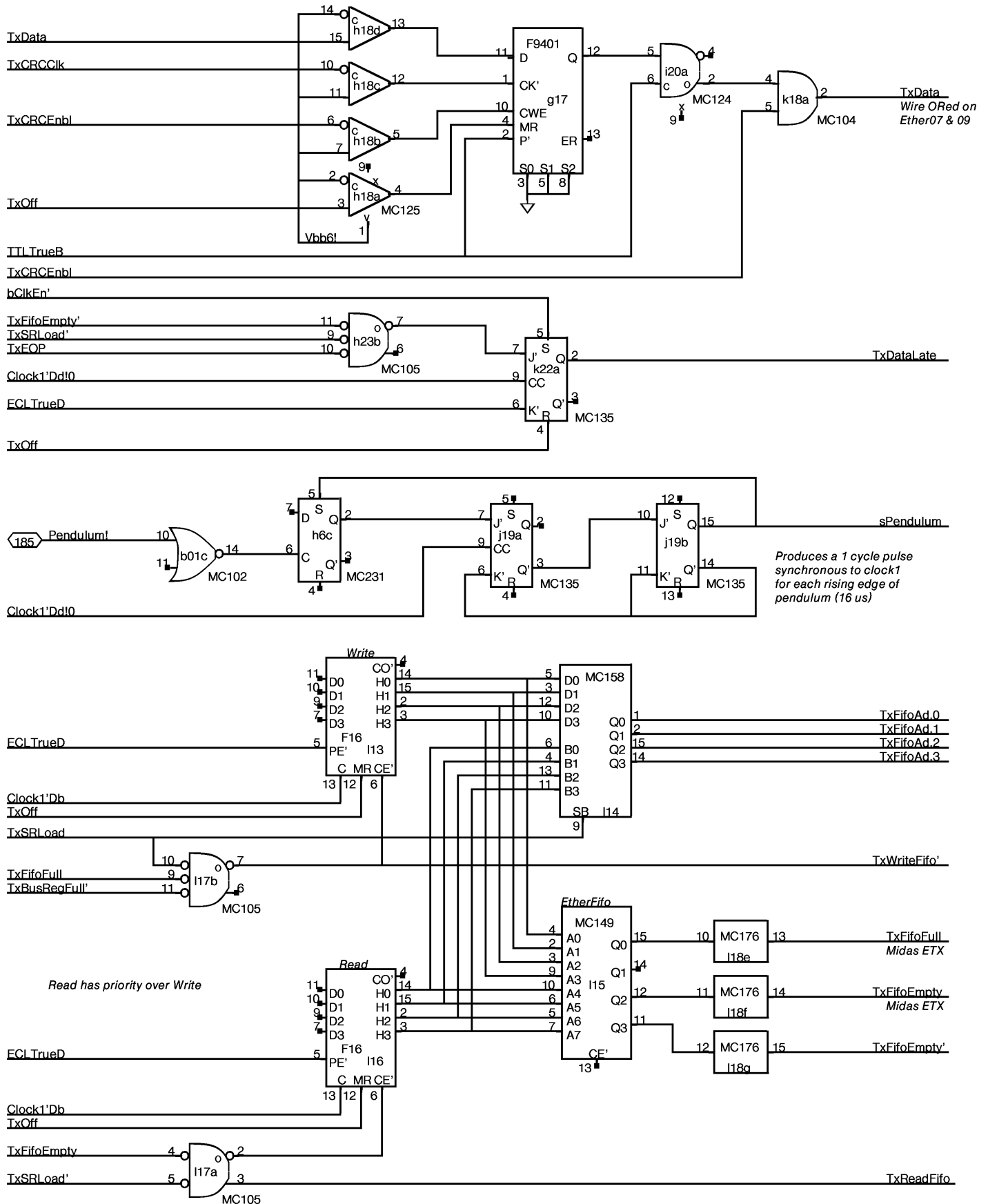


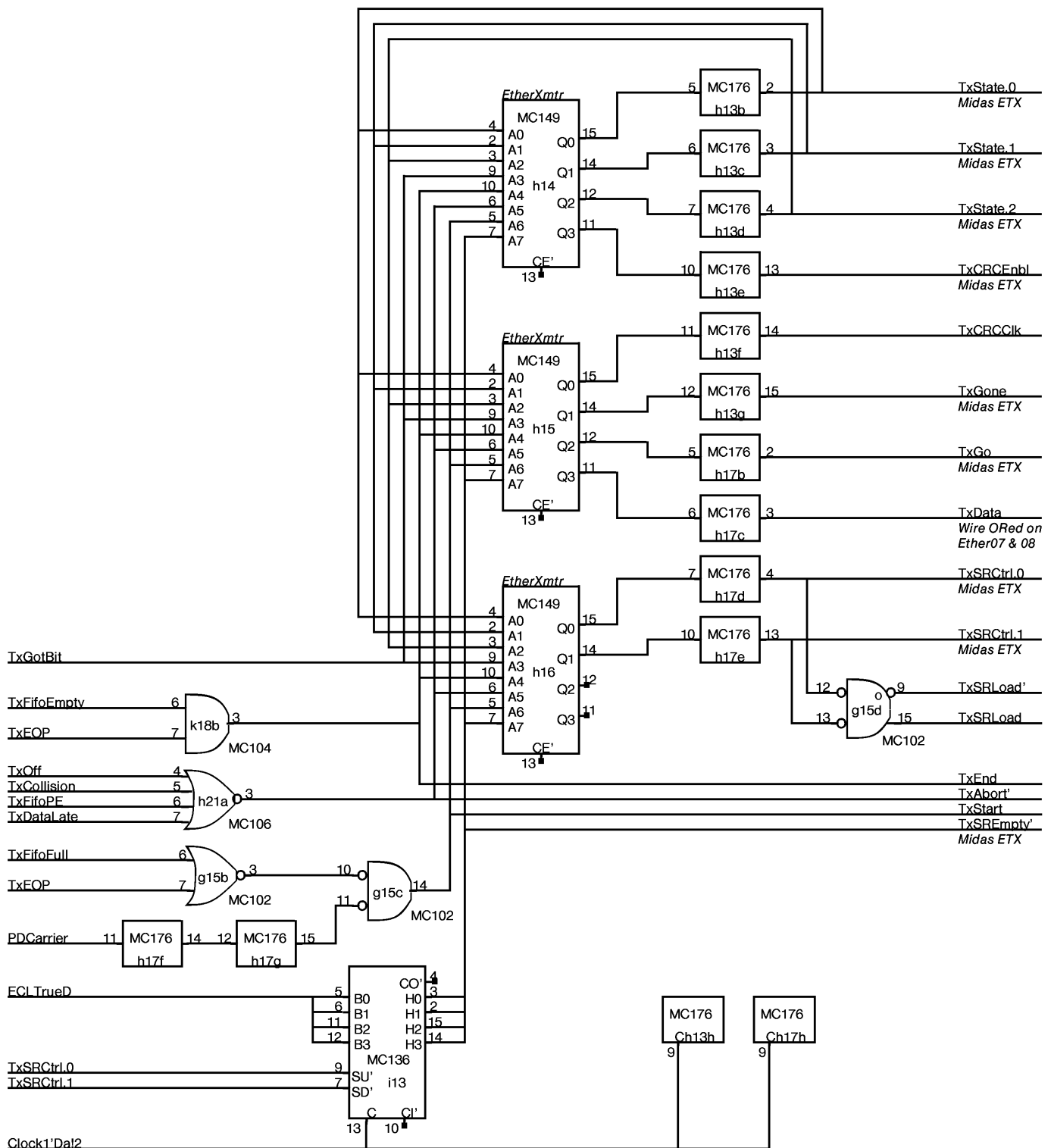
See Ether21 & 22.sil for wakeup timing diagrams



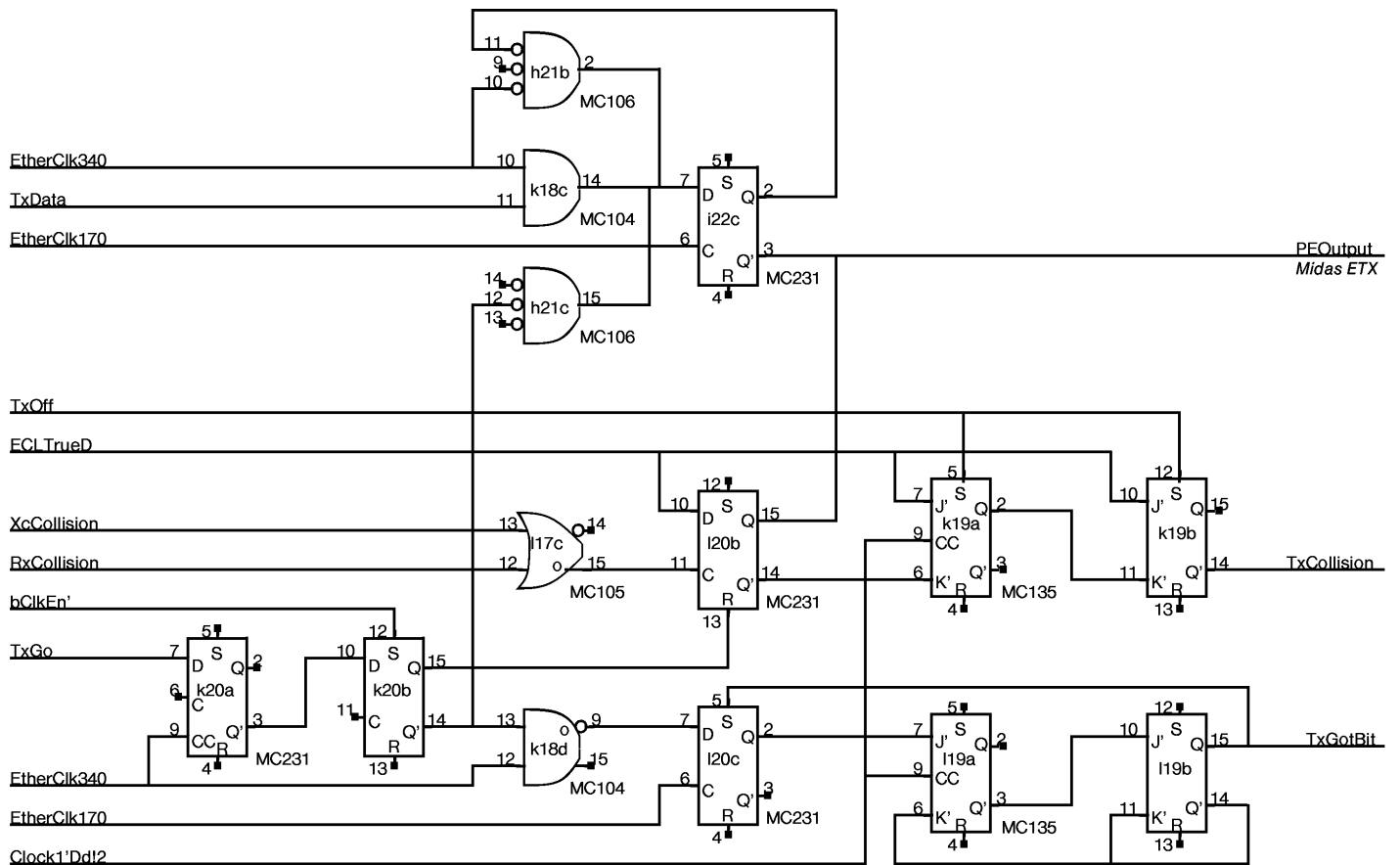
See Ether21 & 22.sil for wakeup timing diagrams







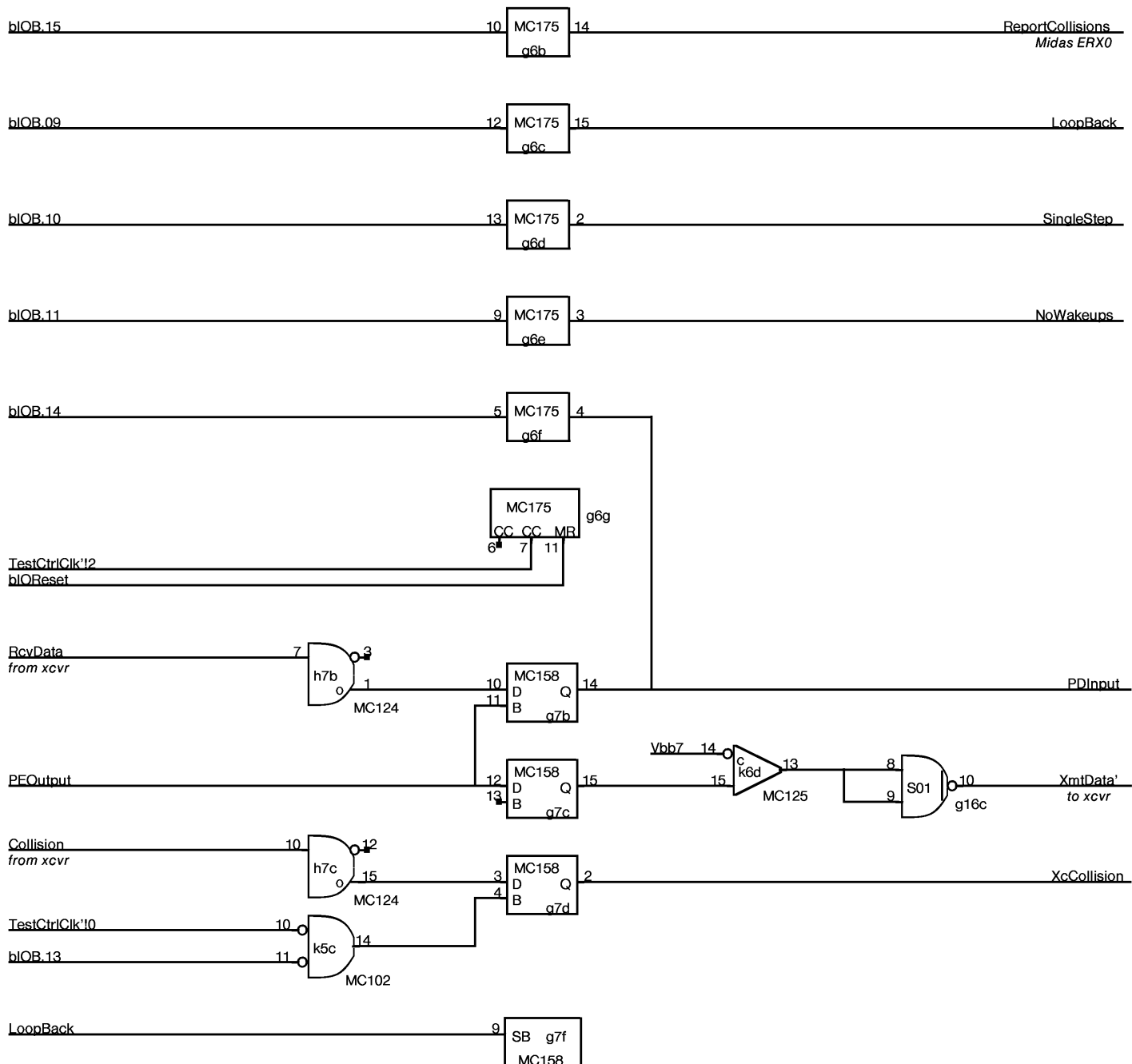
See Ether17.sil for timing diagrams

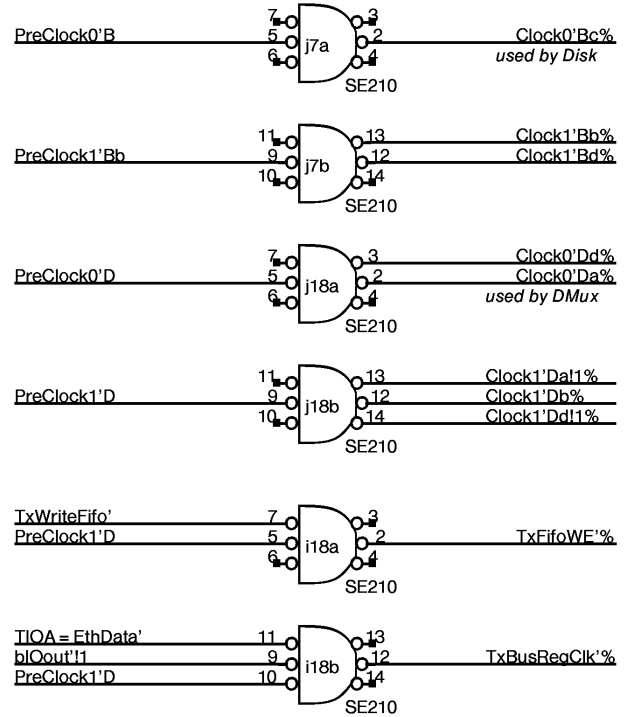
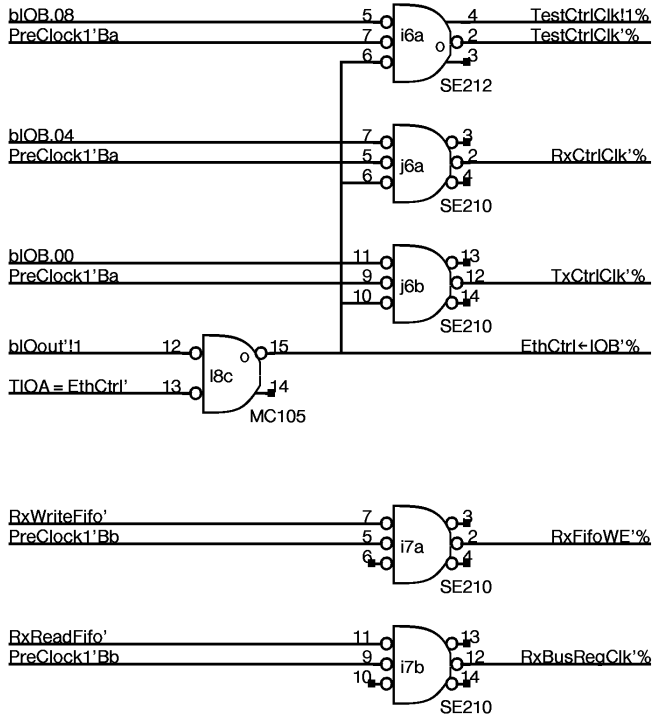


See Ether19.sil for timing diagrams

The slowest Dorado clock speed at which the transmitter works is 42.5 ns (T0 to T1)

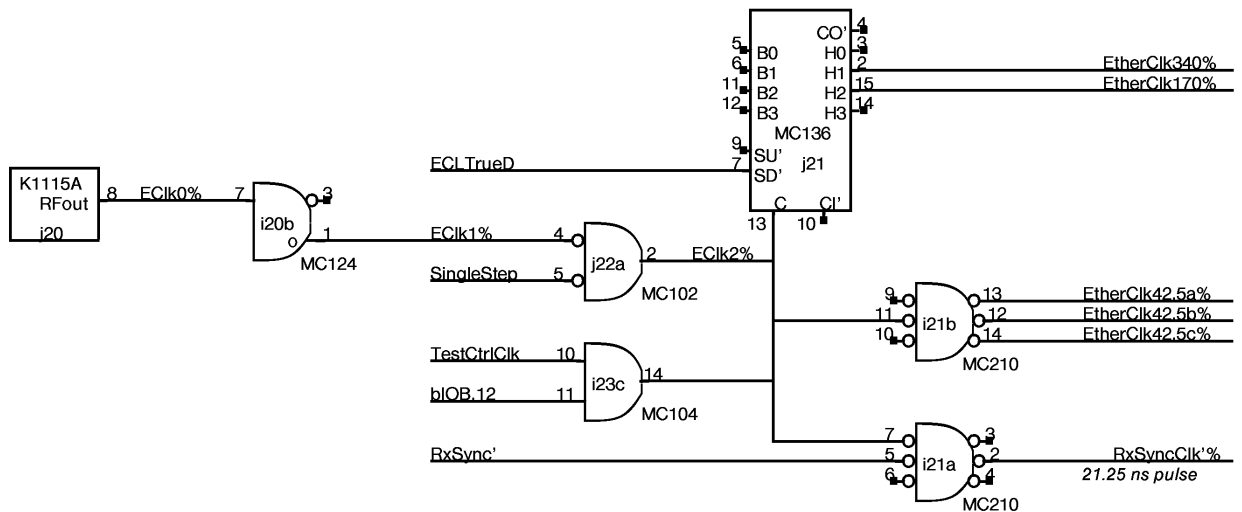
525 ns < length of jam < 915 ns

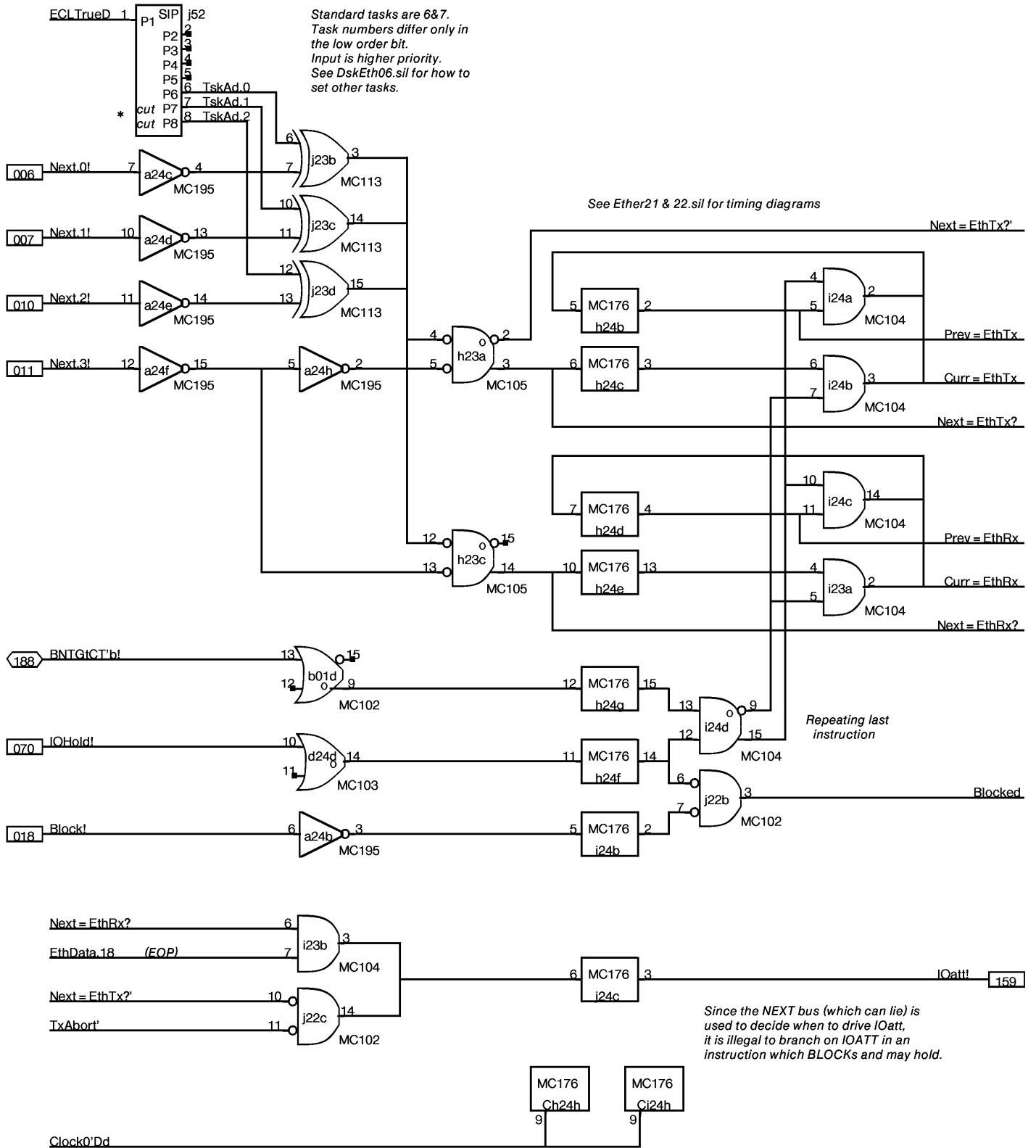


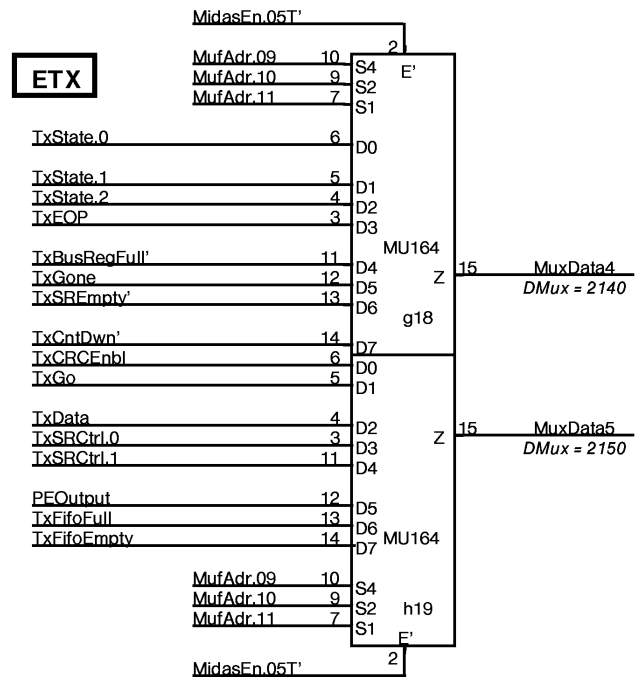
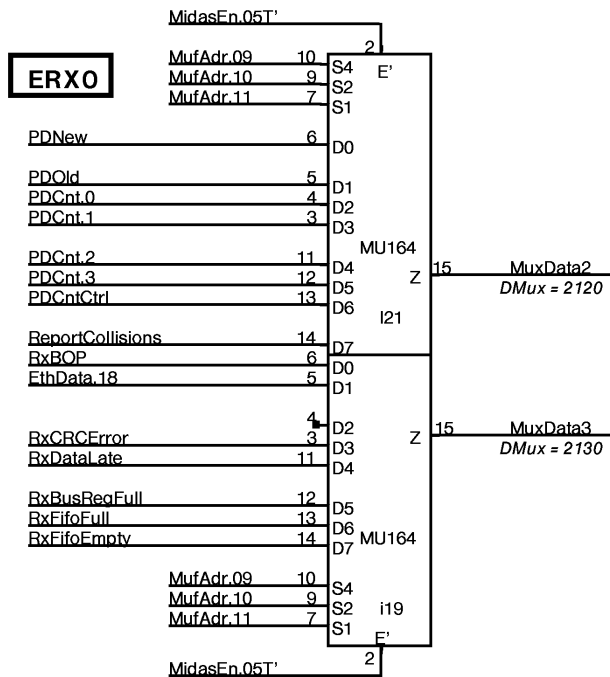


Dorado Synchronous Clocks

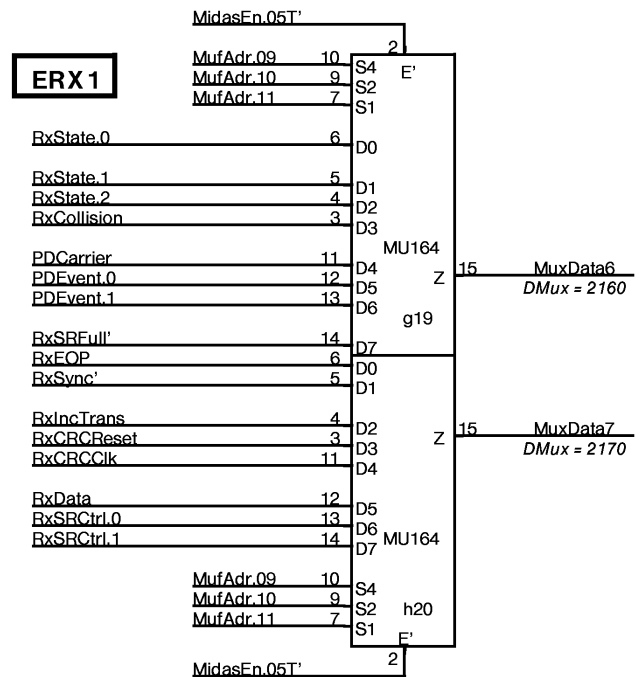
Free-running Ether Clocks





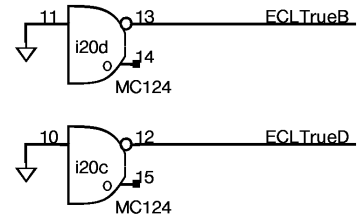
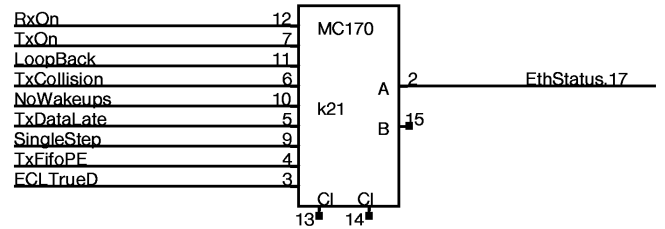
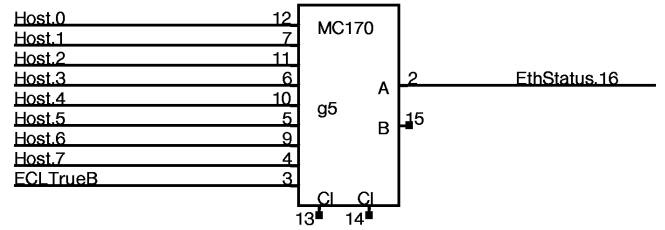
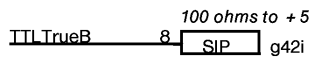
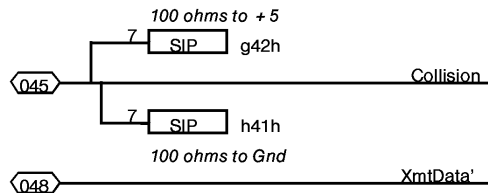
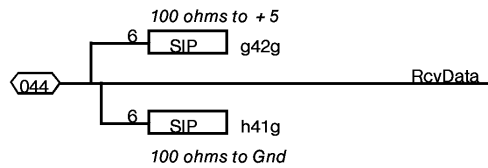


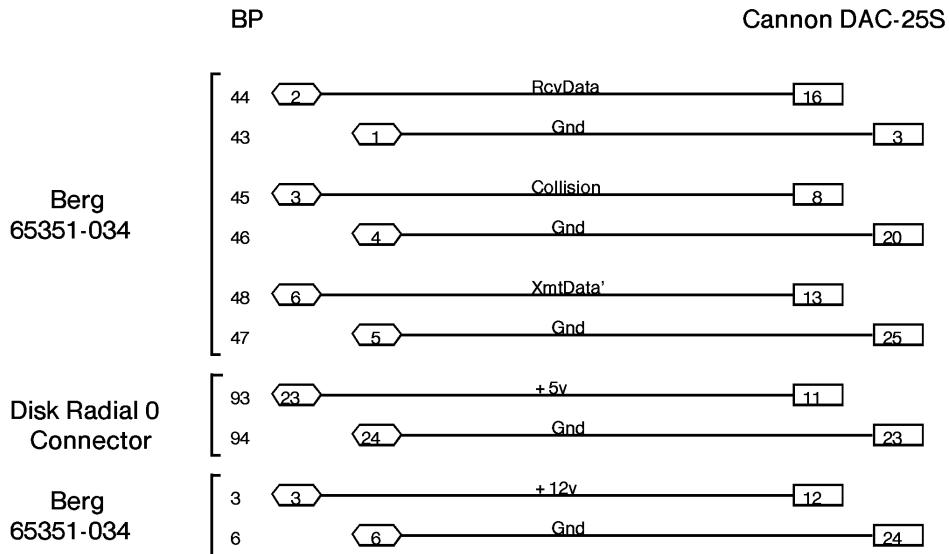
See DskEth01.sil for muffler control logic.
DMux addresses 2000-2117 are used by the disk.



To set a host address bit to 1
pull it up to gnd through 91 ohms.

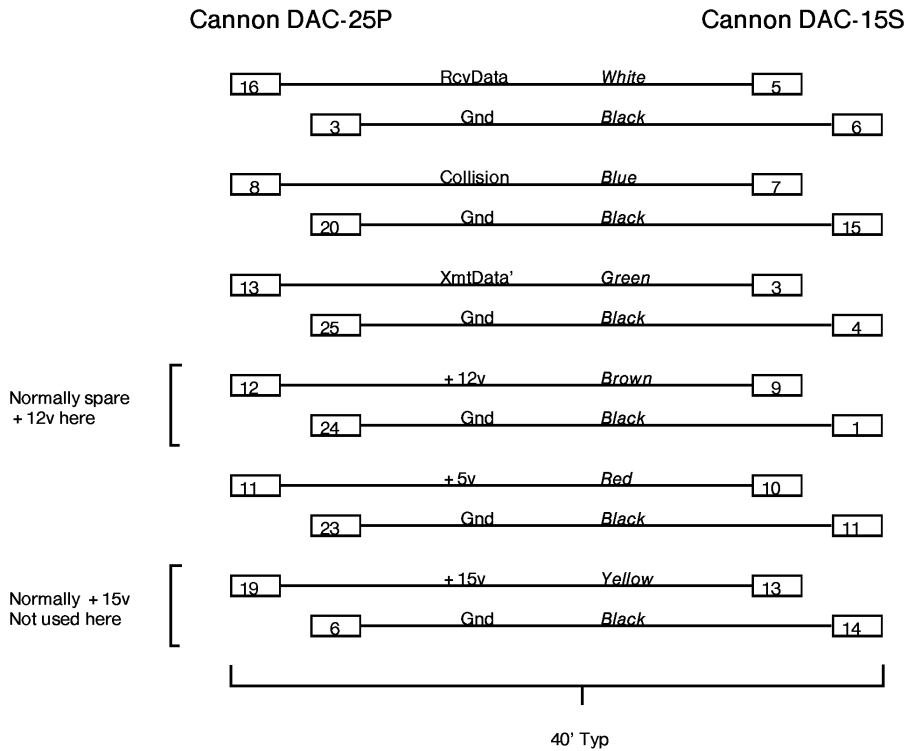
028	Host.0	200
029	Host.1	100
032	Host.2	40
033	Host.3	20
036	Host.4	10
037	Host.5	4
040	Host.6	2
041	Host.7	1



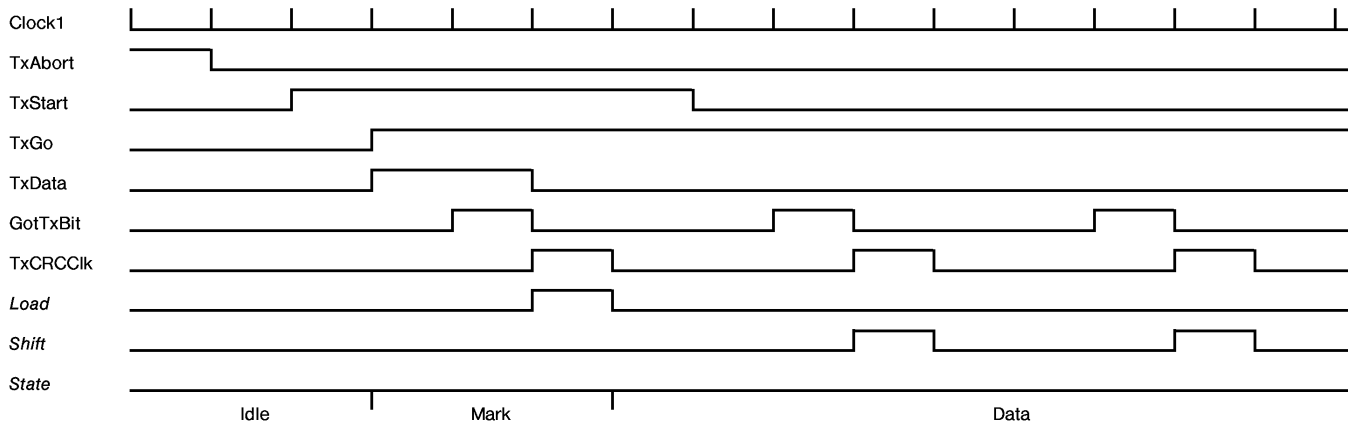


Internal Cable

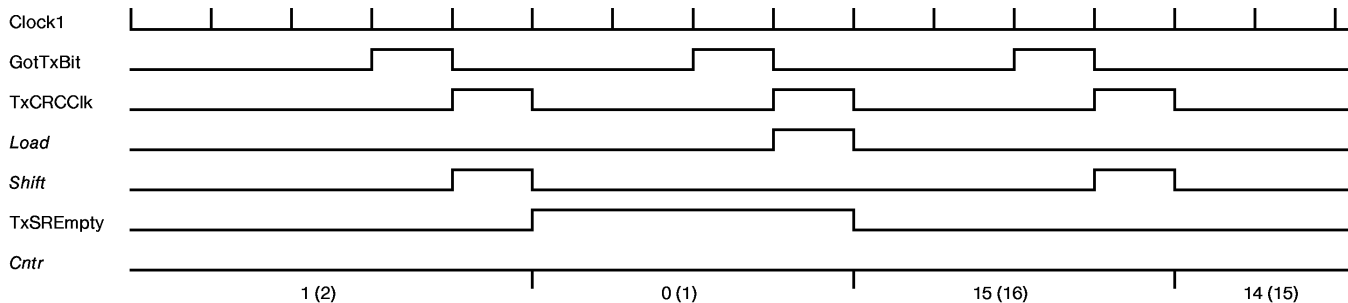
External Cable



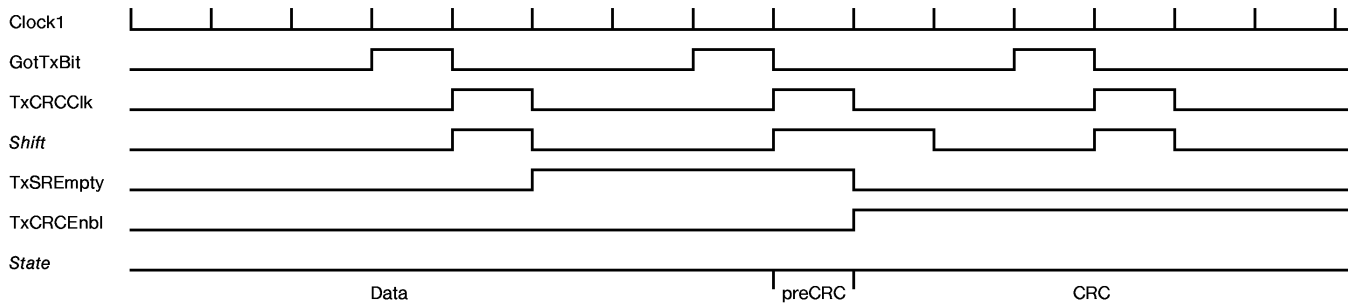
This is a standard Alto II Ethernet external cable part # 216411



Startup

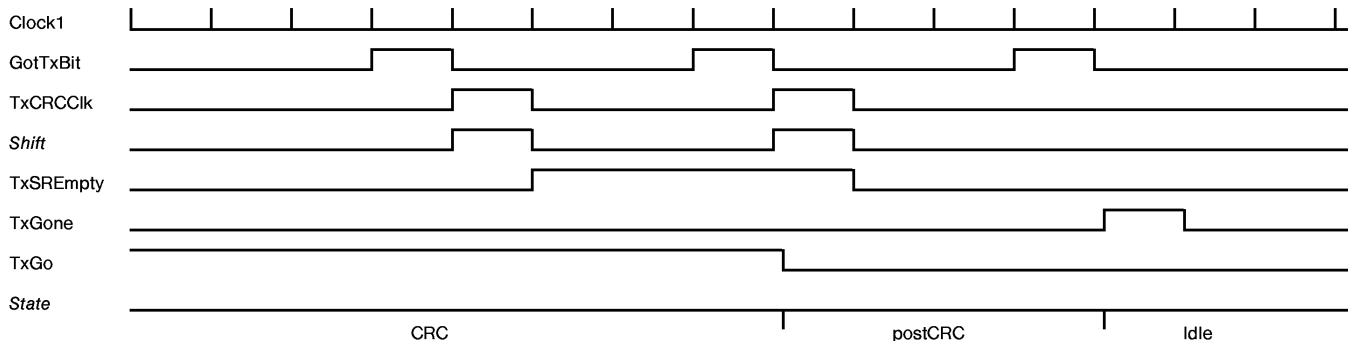


Loading Tx Shift Register

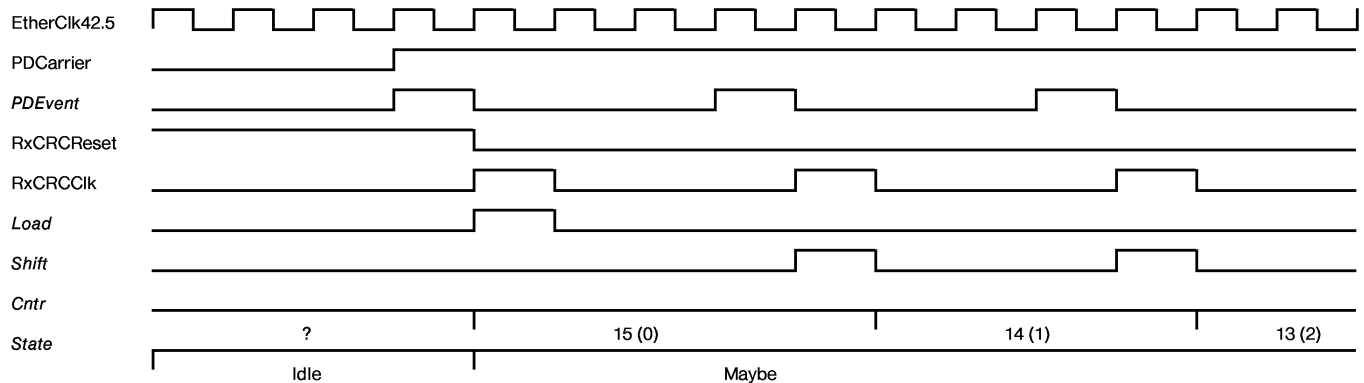


TxEnd has been high since TxSR was last loaded

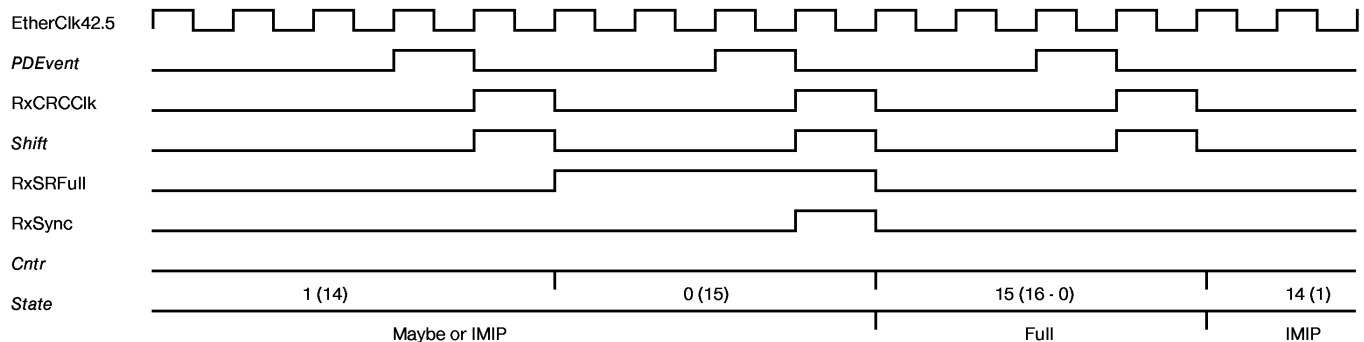
CRC



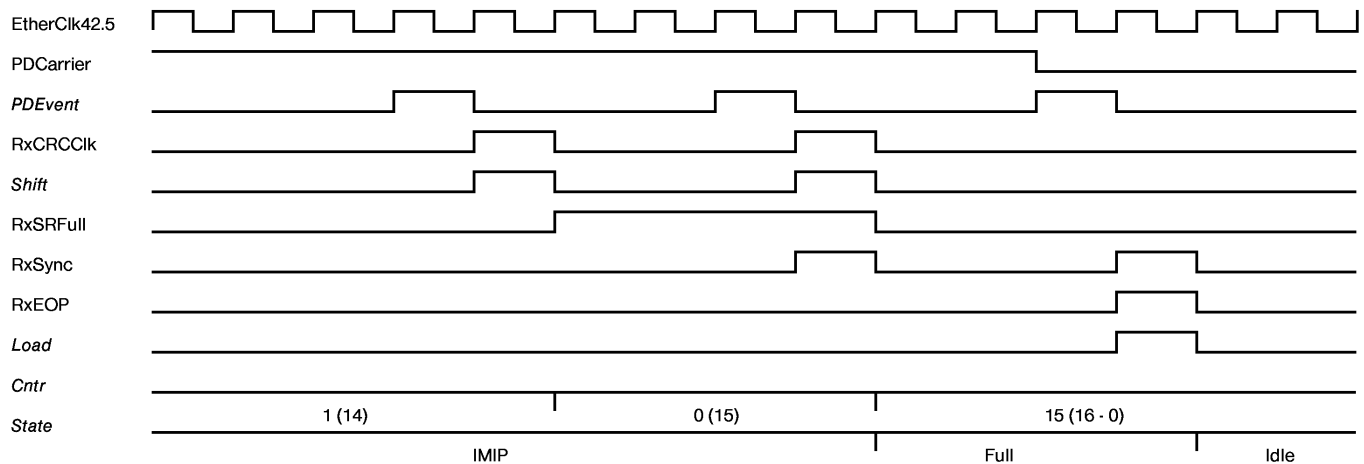
ShutDown



Startup

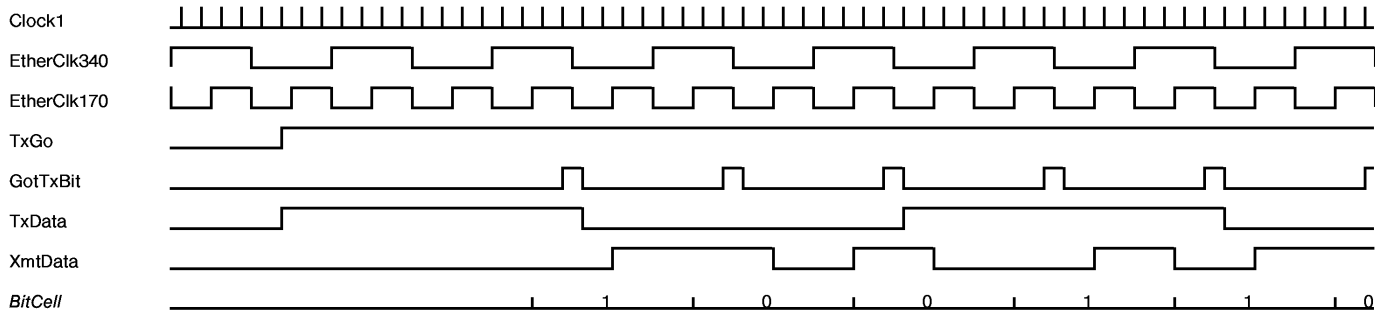


Dumping Rx Shift Register

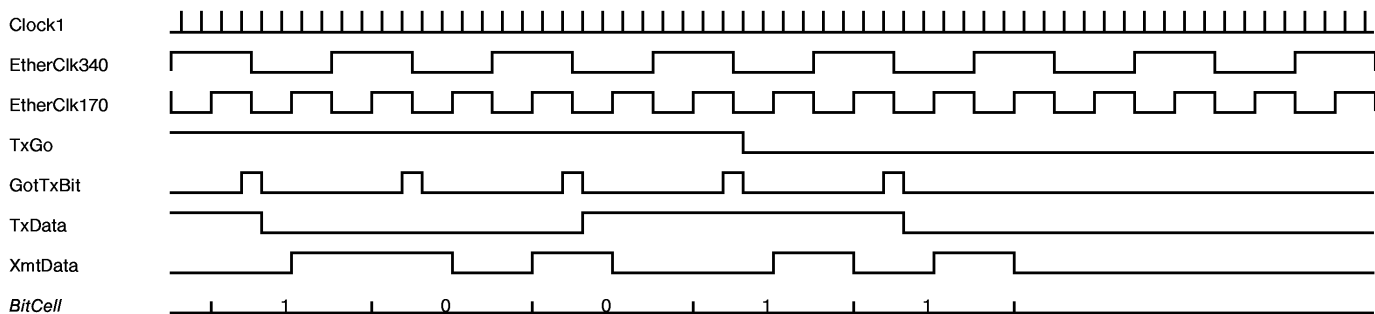


ShutDown

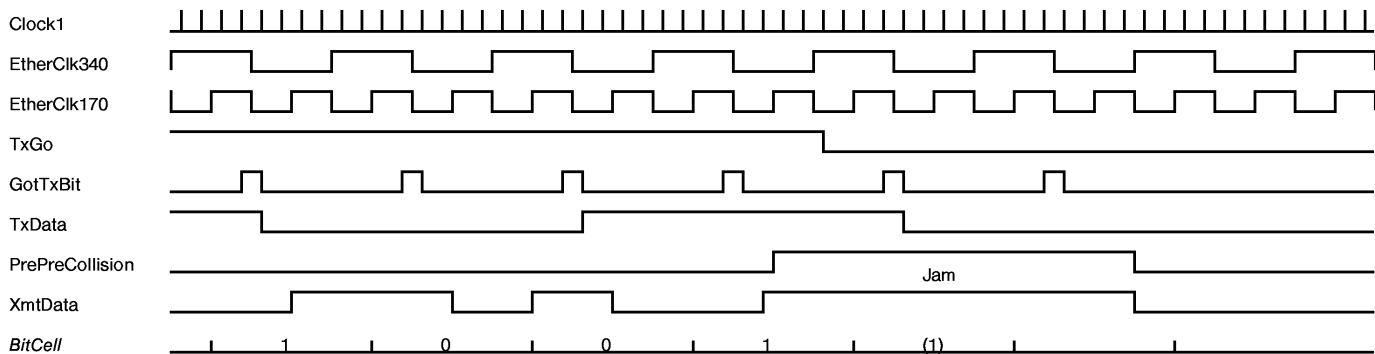
Cntr notation is <cntr value><# occupied postions>
Phase decoder events are encoded in PDEvent.x
Shift and Load are encoded in RxSRCtrl.x
state is encoded in RxState.x
shift also decrements Cntr



Startup

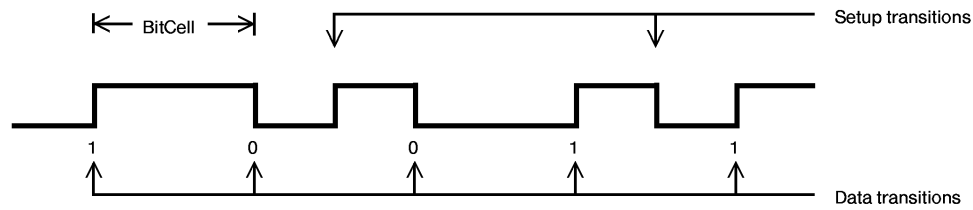


Shutdown

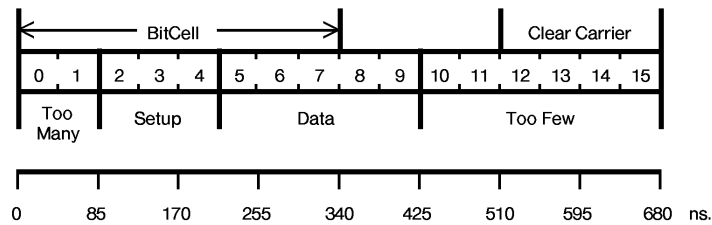


PrePreCollision is the output of the first stage of the Collision synchronizer

Collision

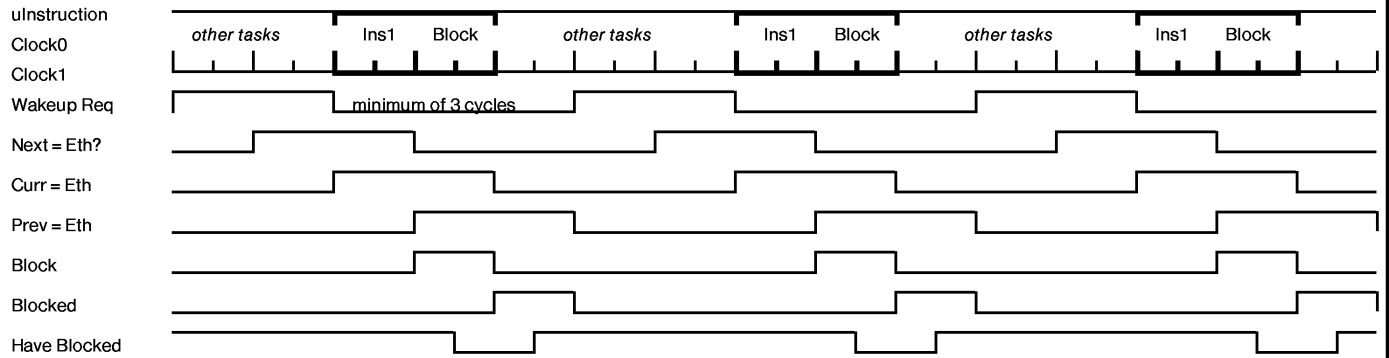


A bitcell is nominally 340 ns.
A sample is nominally 42.5 ns.

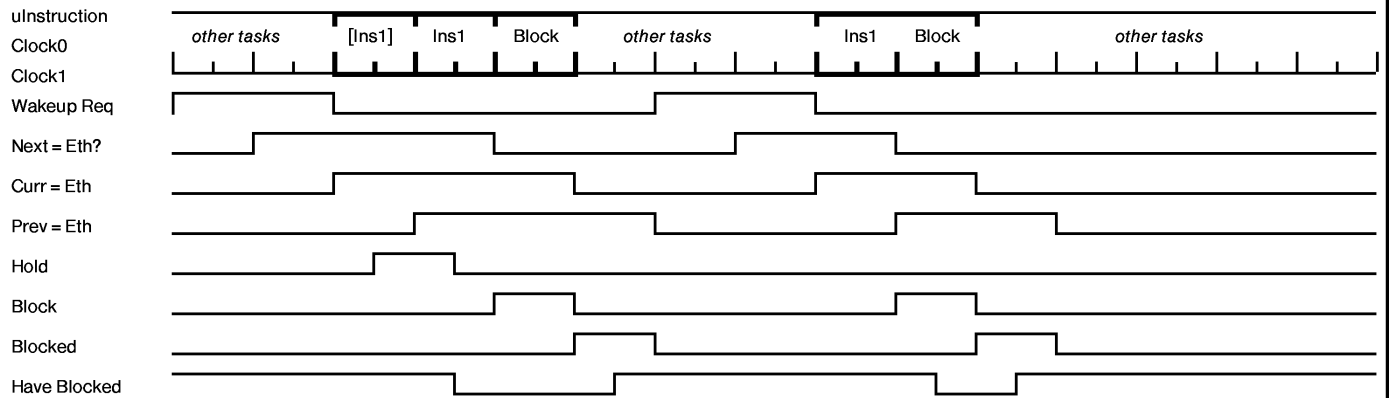


Inputs				Outputs			Comment
Carrier	Old	New	Cnt	Carrier	Event	CntCtrl	
Low	Low	Low	d/c	Low	NoEvent	Count	Idle
Low	Low	High	d/c	High	One	Reset	Start of packet (start bit)
Low	High	Low	d/c	High	Collision	Reset	Impossible
Low	High	High	d/c	Low	NoEvent	Count	Impossible
High	Low	High	0-1	High	Collision	Count	Too many transitions
High	High	Low	0-1	High	Collision	Count	Too many transitions
High	Low	High	2-4	High	NoEvent	Count	Setup transition (zero next)
High	High	Low	2-4	High	NoEvent	Count	Setup transition (one next)
High	Low	High	5-9	High	One	Reset	Data transition
High	High	Low	5-9	High	Zero	Reset	Data transition
High	Low	High	10-15	High	Collision	Reset	Too few transitions
High	High	Low	10-15	High	Collision	Reset	Too few transitions
High	Low	Low	0-11	High	NoEvent	Count	Active
High	High	High	0-11	High	NoEvent	Count	Active
High	Low	Low	12-15	Low	Collision	Reset	End of packet
High	High	High	12-15	High	Collision	Reset	Jam

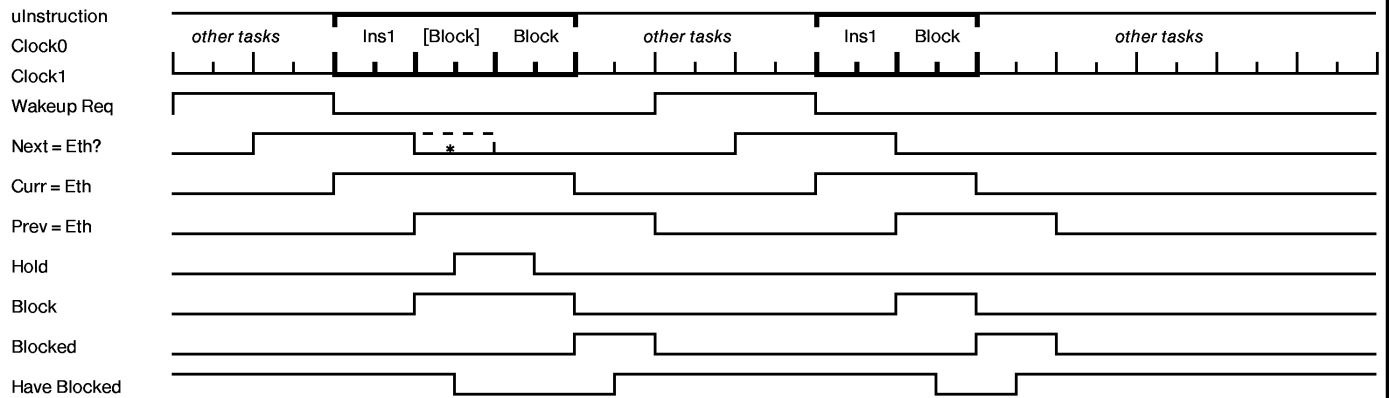
"Impossible" conditions can happen right after power up.
d/c means "don't care".



Normal case

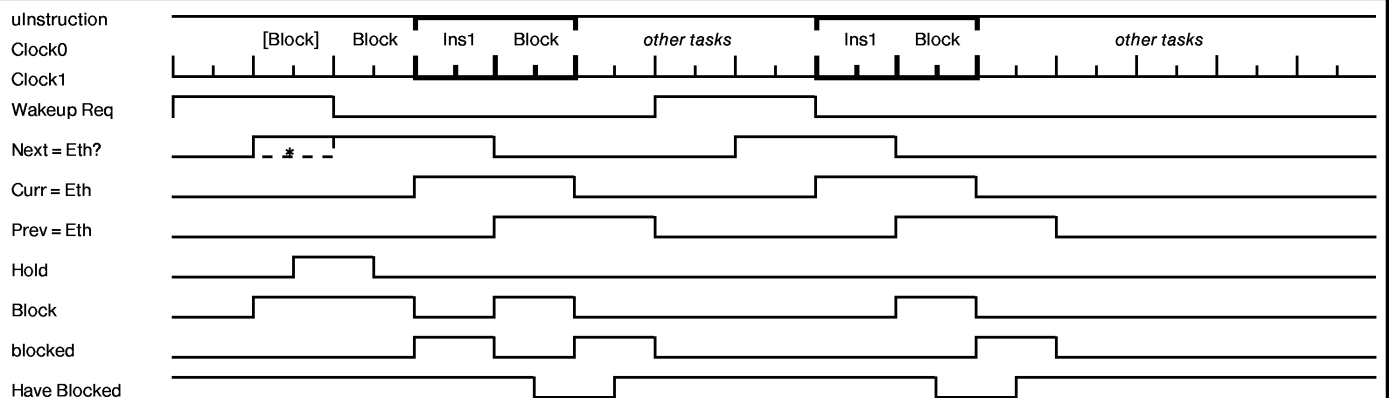


Hold in first instruction & not preempted



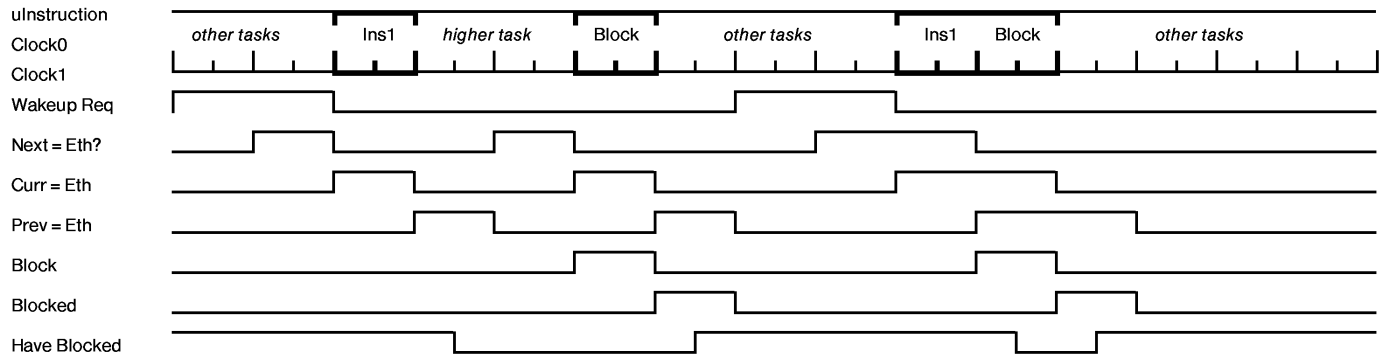
* NEXT bus is wrong

Block & Hold & not preempted

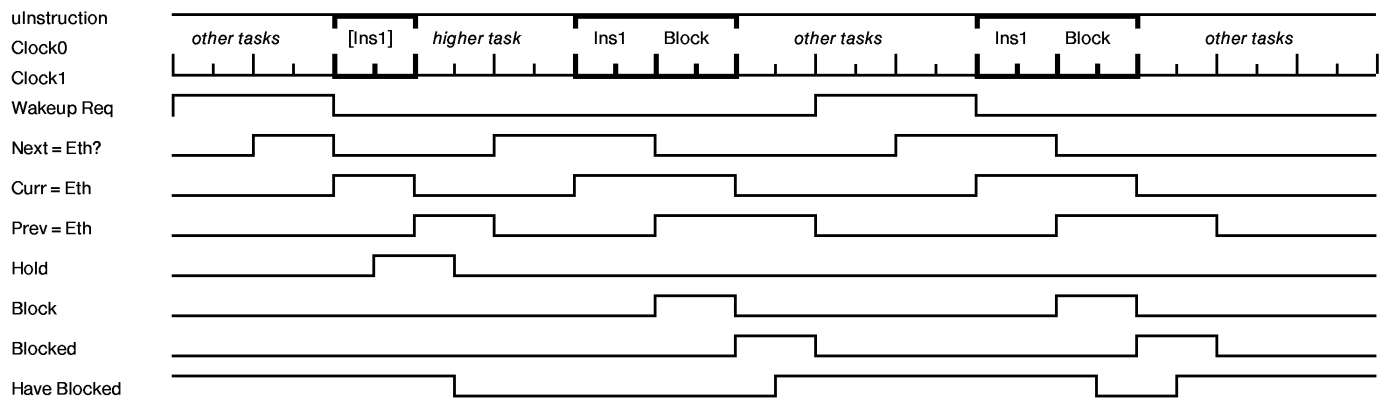


* NEXT bus is wrong

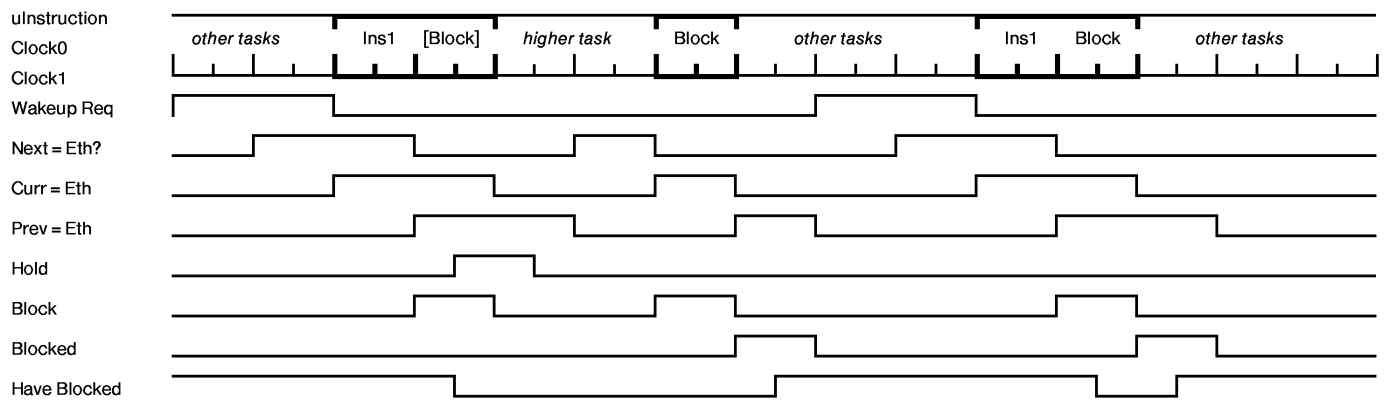
Block & Hold & (BNT = Eth < CTask)



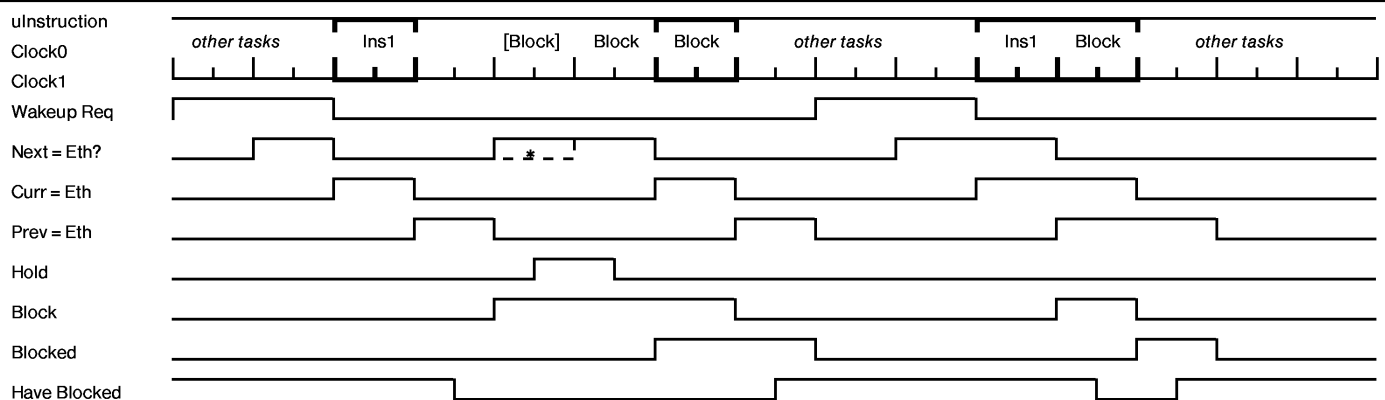
Preemption after first instruction



Hold in first instruction & then preempted



Block & Hold & then preempted



* NEXT bus is wrong

Block & Hold & (BNT = Eth < CTask) while Eth is preempted

00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Tx Cmd Enbl'	Tx On	Tx EOP	Tx Cnt Dwn	Rx Cmd Enbl'	Rx On	Rx BOP'		Test Cmd Enbl'	Loop Back	Single Step	No Wake ups	Test Clock	Test Coll'	Test Data	Report Coils

Output

00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Host Address								Rx On	Tx On	Loop Back	Tx Coll	No Wake ups	Tx Data Late	Single Step	Tx Fifo PE
028	029	032	033	036	037	040	041								

Input

The host address is set by jumpers on the right backplane.
 To set a bit to one, pull it up to ground through 91 ohms.